

On the Elimination of Lateral Forces in Magnetoelastic Rods Confined to Curved Channels

A Thesis

Submitted for the Degree of

Master of Technology

in the Department of Aerospace Engineering

by

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May 2026

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Dedicated to My Parents

DECLARATION

I declare that the thesis entitled **Elimination of Lateral Forces on a Magnetoelastic Rod Confined to Curved Channels** submitted by me for the M. Tech. degree of the Indian Institute of Science did not form the subject matter of any other thesis submitted by me for any outside degree. The work incorporated in this thesis is original and carried out by me at the Indian Institute of Science, Bengaluru, India.

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Acknowledgment

I would like to express my deepest gratitude to the numerous individuals who have contributed to the successful completion of my M. Tech. thesis. Their unwavering support, guidance, and encouragement have been invaluable throughout this journey. First and foremost, I want to express my sincere gratitude to my research adviser, Prof. Vivekanand Dabade. His expertise, insightful and honest feedback, and unwavering dedication have been instrumental in shaping the direction of this thesis.

I am also deeply grateful to my labmate, Mohd Tahseen, PhD scholar, for the many invaluable discussions, generous guidance, and patient clarification of doubts throughout this work. His inputs have greatly enriched my understanding and have contributed to this thesis.

Notations and Nomenclature

Notation	Description
s	arc length parameter
$\mathbf{r}(s)$	position vector of the rod centerline
$\theta(s)$	angle between cross-section area and $\mathbf{e}_3$
$\kappa(s)$	curvature of the rod
$\mathbf{m}(s)$	magnetization vector
M_s	saturation magnetization
$\gamma(s)$	angle between $\mathbf{m}(s)$ and $\mathbf{e}_3$ in the reference configuration (magnetization angle)
$\mathbf{H}_{ext}$	external magnetic field vector
H_{ext}	magnitude of external magnetic field
β	angle between $\mathbf{H}_{ext}$ and $\mathbf{e}_3$
$\tau(s)$	magnetic torque
$\mathbf{F}_{ext}$	external force acting on the left end of the rod
$\mathbf{e}_3, \mathbf{e}_1, \mathbf{e}_2$	reference configuration axes
$\mathbf{d}_3, \mathbf{d}_1, \mathbf{d}_2$	deformed configuration axes
$\nu(s)$	stretch
$\eta(s)$	shear strain
$N(s)$	internal axial force
$H(s)$	internal shear force
$M(s)$	internal moment
EI	bending stiffness of the rod
A	area of cross-section of the rod
$\phi_\Gamma(s)$	graph of Γ
$\Gamma(s)$	centerline of the channel
$\Theta(s)$	angle made between tangent vector to $\Gamma(s)$ and $\mathbf{e}_3$
$k(s)$	curvature of the channel
s_0	left end of the rod inside the channel
s_L	right end of the rod inside the channel

Abstract

Flexible slender rods navigating confined curved pathways arise in applications such as minimally invasive medical procedures, catheter guidance, and continuum robotics, where contact with surrounding boundaries generate undesirable lateral contact forces. This thesis investigates the mechanics of a planar magnetoelastic Cosserat rod constrained within prescribed curved channels, with the objective of eliminating lateral contact forces through distributed magnetic actuation.

For channels with continuous curvature, an inverse formulation is developed to determine the magnetic torque distribution required to eliminate distributed lateral forces while maintaining equilibrium for a prescribed rod shape. A practically relevant constant-magnetization strategy is also examined, leading to a geometric admissibility condition for channel curvatures that allow quasi-static rod insertion.

The formulation is extended to channels with discontinuous curvature, where curvature jumps create singular interface effects and corresponding jump conditions in internal forces and bending moment. It is shown that magnetic actuation can remove both distributed lateral forces and concentrated point forces, although concentrated moments arising from curvature discontinuities remain in the present formulation. A constant-magnetization insertion analysis is also developed for piecewise-curvature channels, yielding an admissibility condition similar to that of the continuous-curvature case.

A direct formulation is further developed to validate the inverse approach and to explore a broader class of equilibrium configurations arising from different boundary conditions, beyond those accessible through the inverse framework. Overall, the results establish an analytical framework for magnetically actuated constrained rod systems, with applications in confined insertion and robotic navigation.

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Chapter 1

Introduction

1.1 Background and Motivation

Slender structures undergoing large deformations are encountered in a wide range of engineering and biological systems. Their behavior is governed by a strong coupling between geometry and mechanics, requiring models that remain valid under large rotations and nonlinear kinematics. Cosserat rod theory provides a geometrically exact framework for describing such structures by accounting for bending, twisting, and stretching through the balance of forces and moments along the rod centerline.

A class of problems of particular interest involves rods constrained to lie within prescribed geometries, such as channels or tubes. A prominent motivating example is colonoscopy, where a flexible instrument must be advanced through the highly curved and tortuous geometry of the large intestine while minimizing contact forces on the surrounding tissue. Related scenarios also arise in flexible medical devices, soft robotic manipulators, and confined insertion mechanisms. In such cases, the rod is required to conform to a given path while maintaining mechanical equilibrium.

However, when an elastic rod is forced to follow an arbitrary channel geometry, internal stresses are generated that typically result in non-zero action-reaction force pairs due to contact with the surrounding channel [1]. These forces can impede smooth motion, increase insertion resistance, and lead to undesirable interactions with the environment. Consequently, achieving controlled deformation of rods within confined geometries remains a challenging problem.

1.2 Magnetoelastic Actuation

Magnetoelastic rods offer a promising approach to overcoming these limitations. By embedding magnetic properties within the rod and applying an external magnetic field, distributed torques can be generated along its length. These torques provide a mechanism for remotely controlling the deformation of the rod without requiring direct mechanical actuation.

The key idea explored in this work is that magnetic torque can be used to counterbalance the elastic forces that arise when a rod is constrained to follow a prescribed geometry. If appropriately designed, the magnetic actuation can eliminate lateral forces, enabling the rod to conform to the desired path without exerting any force on the channel walls. This capability is particularly relevant for applications requiring smooth and controlled insertion in constrained environments.

1.3 Problem Statement

The problem considered in this thesis is the equilibrium of a slender rod constrained to lie within a prescribed channel geometry. The channel is characterized by a curvature distribution, and the rod is required to conform to this geometry while satisfying mechanical equilibrium.

In the absence of external actuation, a purely elastic rod generally develops non-zero lateral forces when constrained to follow arbitrary paths. The objective of this work is to determine whether magnetic actuation can be used to eliminate these lateral forces and, if so, to identify the conditions under which this is possible.

Two classes of channel geometries are considered. In the first case, the channel curvature varies continuously along the length of the rod. In the second case, the curvature exhibits discontinuities, leading to jumps in internal forces and moments [2]. The analysis addresses both scenarios and investigates the role of magnetic torque in achieving force-free configurations.

1.4 Objectives and Contributions

The primary objective of this work is to develop a systematic framework for analyzing magnetoelastic rods constrained within prescribed channel geometries and to determine conditions under which lateral forces can be eliminated.

The key contributions of this thesis are as follows:

- Formulation of the equilibrium equations for rods constrained to follow prescribed

channel geometries using Cosserat rod theory [1].

- Extension of the formulation to include magnetic actuation through distributed torques.
- Derivation of conditions under which lateral forces acting on the channel vanish.
- Analytical solutions for the inverse problem, in which the required magnetic actuation is determined for a given channel geometry.
- Formulation and solution of the direct problem, where the resulting rod configuration is obtained from prescribed loads and magnetic torque.
- Treatment of both continuous and discontinuous curvature profiles, including the derivation of jump conditions for internal forces and moments.
- Identification of special solutions corresponding to a constant magnetization angle.
- Illustrative examples based on canonical geometries.

1.5 Modeling Assumptions

The formulation developed in this work is based on the following assumptions:

- **Planar, inextensible, and unshearable rod:** The rod is assumed to deform in a plane. It is modeled using the Cosserat formulation, and the behavior of the material is assumed to be linearly elastic. We impose the constraints that the rod is inextensible, unshearable, and twist-free.
- **Rigid, planar, and frictionless channel:** The channel geometry is rigid, wherever prescribed, and does not deform under loading. The interaction between the rod and the channel is assumed to be frictionless. The contact between the rod and the channel is treated through the constraint that the rod follows the channel path.
- **Quasi-static equilibrium:** Dynamic effects such as inertia are neglected, and the rod is assumed to be in static equilibrium under the applied loads.
- **Boundary loading:** An external force $\mathbf{F}_{\text{ext}}$ is applied at the boundary (left end) of the rod. No distributed external applied forces are considered.

- **Uniform applied magnetic field:** Magnetic actuation is incorporated through a distributed torque that depends on the orientation of the rod with respect to an applied magnetic field, which is uniform.

Unless otherwise specified, all physical quantities used throughout this thesis are expressed in SI units.

1.6 Organization of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 presents the Cosserat rod formulation and the notation used throughout the thesis, including rod kinematics, balance laws, constitutive relations, and magnetic actuation. The distinction between continuous and discontinuous curvature channels is also introduced.

Chapter 3 investigates the behavior of elastic and magnetoelastic rods in channels with continuous curvature. Expressions for internal forces and lateral contact force are derived, conditions for elimination of the lateral force are established, and analytical solutions for several canonical channel geometries are obtained. The chapter further examines constant magnetization angle solutions relevant to rod insertion.

Chapter 4 extends the formulation to channels with discontinuous curvature. Jump conditions associated with curvature discontinuities are derived, and the resulting concentrated force and moment effects are analyzed. Conditions for lateral-force elimination in piecewise-curved channels are subsequently established, followed by rod insertion solutions for several discontinuous-curvature geometries, including intestinal-type pathways.

Chapter 5 addresses the direct problem for both continuous and discontinuous curvature channels. Using the loading and magnetic actuation obtained from the inverse formulation, the chapter demonstrates that the prescribed equilibrium configurations can be reproduced through appropriate boundary loads and magnetization distributions.

Finally, Chapter 6 summarizes the principal findings of the present work, discusses the key physical and mathematical insights obtained from the formulation, and outlines possible directions for future research.

Chapter 2

Cosserat Rod Formulation and Problem Setup

2.1 Cosserat Rod Kinematics

A slender rod is modeled as a one-dimensional continuum characterized by its centerline and orientation. The centerline of the rod is described by a vector-valued function $\mathbf{r}(s)$, where $s \in [0, L]$ denotes the arc length parameter.

At each point along the rod, an orthonormal director frame $\{\mathbf{d}_1(s), \mathbf{d}_2(s), \mathbf{d}_3(s)\}$ is attached, where $\mathbf{d}_1(s)$ and $\mathbf{d}_2$ lie in the plane of its deformed cross sections and $\mathbf{d}_3(s) = \mathbf{d}_2(s) \times \mathbf{d}_1(s)$.

We define strain variables ν and η through the following decomposition along the director frame:

$$\mathbf{r}_s(s) = \nu \mathbf{d}_3(s) + \eta \mathbf{d}_1(s),$$

where the subscript s denotes the partial derivative with respect to the arc-length parameter. The function $\nu = \nu(s)$ describes the stretch while $\eta = \eta(s)$ defines the shear strain.

The orientation of the deformed cross-section is obtained through the function $\theta(s)$ defined by

$$\mathbf{d}_3(s) = \cos \theta(s) \mathbf{e}_3 + \sin \theta(s) \mathbf{e}_1.$$

Thus, we have

$$\mathbf{d}_1(s) = -\sin \theta(s) \mathbf{e}_3 + \cos \theta(s) \mathbf{e}_1.$$

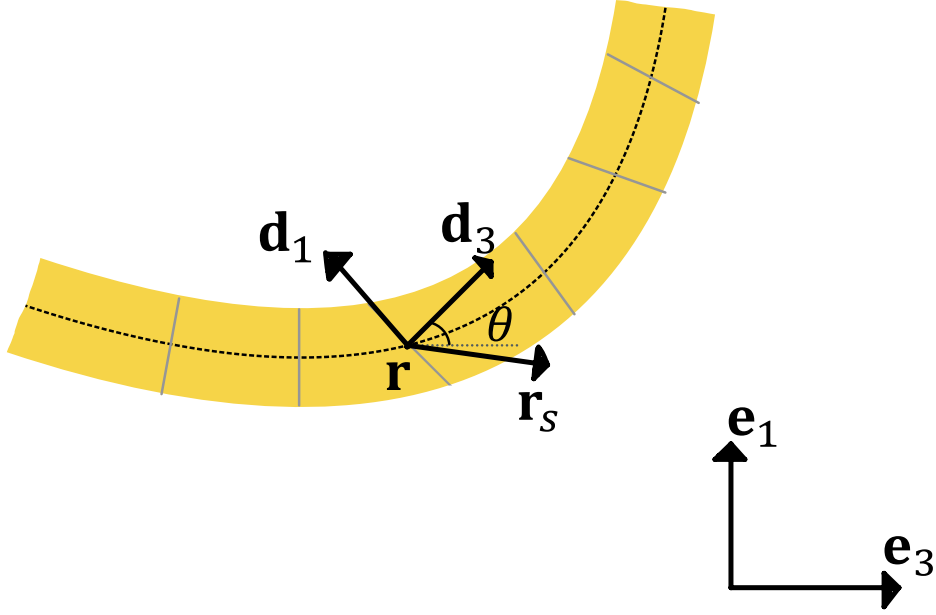


Figure 2.1: Planar Cosserat rod configuration showing the centerline position vector $\mathbf{r}$, local director basis $(\mathbf{d}_1, \mathbf{d}_3)$, tangent vector $\mathbf{r}_s$, and tangent angle θ measured relative to the fixed basis $(\mathbf{e}_1, \mathbf{e}_3)$. Grey segments denote rod cross sections.

The constraints of inextensibility and unshearability imply:

$$\nu(s) = 1 \quad \text{and} \quad \eta(s) = 0.$$

2.2 Balance Laws

The equilibrium of the rod is governed by the balance of internal forces and moments. Let $\mathbf{n}(s)$ denote the internal force resultant and $\mathbf{M}(s)$ denote the internal moment resultant.

The balance of linear momentum is expressed as

$$\mathbf{n}_s(s) + \mathbf{f}(s) = \mathbf{0},$$

where $\mathbf{f}(s)$ represents the distributed external force per unit length.

The balance of angular momentum is given by

$$\mathbf{M}_s(s) + \mathbf{r}_s(s) \times \mathbf{n}(s) + \mathbf{M}_{ext}(s) = \mathbf{0},$$

where $\mathbf{M}_{ext}(s)$ represents the distributed external moment per unit length[3].

These equations form the fundamental governing equations for the Cosserat rod

and will be specialized in subsequent chapters to account for kinematic and geometric constraints and magnetic actuation.

2.3 Constitutive Relation

The internal moment $M(s)$ is related to the deformation of the rod through a constitutive relation. For a linearly elastic rod undergoing planar deformations, the moment is typically expressed as a function of curvature:

$$M(s) = EI\kappa(s),$$

where

$$\kappa(s) = \theta_s(s).$$

EI is the bending stiffness of the rod, $\kappa(s)$ represents the curvature of the rod and $\theta_s(s)$ gives the bending strain. This relation provides a link between the geometric deformation of the rod and the internal moment distribution.

2.4 Internal Force Components

The internal force $\mathbf{n}(s)$ is decomposed into components aligned with the director frame. In particular, the axial force $N(s)$ and shear force $H(s)$ are defined as

$$N(s) = \mathbf{n}(s) \cdot \mathbf{d}_3(s), \quad H(s) = \mathbf{n}(s) \cdot \mathbf{d}_1(s).$$

The internal forces N and H enforce the constraints of inextensibility and unshearability, respectively [1].

2.5 External Force

The rod is assumed to be partially or completely inserted into a prescribed channel. Let s_0 and s_L denote the arc-length coordinates corresponding to the left and right ends of the rod segment lying inside the channel, respectively.

A external force $\mathbf{F}_{ext}$ is applied at the left end of the rod to hold the rod in equilibrium during insertion. We assume the external force to be along the tangent of the channel. The point corresponding to $s = s_0$ therefore represents the rod end near the channel entry, while $s = s_L$ denotes the distant end of the rod inside the channel.

As the rod is inserted further into the channel, the values of s_0 and s_L may vary while

satisfying

$$s_L - s_0 = L,$$

where L denotes the length of the rod segment constrained within the channel.

The external force $\mathbf{F}_{ext}$ plays a central role in balancing the internal elastic and magnetic forces generated during deformation and insertion.

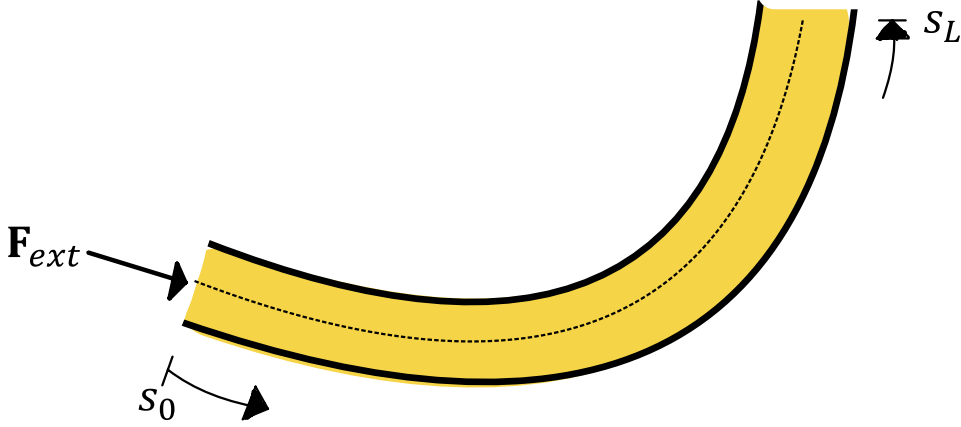


Figure 2.2: Rod segment constrained within a prescribed channel geometry. The coordinates s_0 and s_L denote the left and right boundaries of the rod segment inside the channel, respectively, while the constant external force $\mathbf{F}_{ext}$ maintains equilibrium at the insertion boundary.

2.6 Rod in a Prescribed Channel

The rod is assumed to be constrained to lie within a prescribed channel geometry. We impose the constraint:

$$\mathbf{r} \in \text{Graph}\{\boldsymbol{\Gamma}\} \quad \text{or} \quad \phi_{\boldsymbol{\Gamma}}(\mathbf{r}) = 0,$$

where the equation $\phi_{\boldsymbol{\Gamma}} = 0$ defines the curve $\boldsymbol{\Gamma}$, which is interpreted as the centerline of the channel. The tangent to the centerline is denoted by $\boldsymbol{\Gamma}_s(s)$, where $\Theta(s)$ is the angle made by the tangent with respect to $\mathbf{e}_3$. The curvature of the centerline is denoted by $\kappa_c(s)$.

We assume $|\nabla\phi_{\boldsymbol{\Gamma}}| = 1$. The channel constraint is imposed by the lateral force f [1].

2.7 Magnetic Actuation

Magnetic actuation is incorporated through a distributed torque, $\boldsymbol{\tau}(s)$ applied along the rod. It denotes the magnetic torque per unit length, which depends on the magnetization

of the rod and the applied magnetic field.

We define the magnetization vector $\mathbf{m}(s)$ in the deformed configuration as:

$$\mathbf{m}(s) = M_{sat} \hat{\mathbf{m}}(s),$$

where

$$\hat{\mathbf{m}}(s) = \cos[\theta(s) + \gamma(s)] \mathbf{e}_3 + \sin[\theta(s) + \gamma(s)] \mathbf{e}_1.$$

M_{sat} is the saturation magnetization and $\gamma(s)$ represents the orientation of the magnetization vector with respect to the normal of the cross-section, i.e., $\mathbf{e}_3$ in the reference configuration. The magnetization vector is attached rigidly to the cross-section of the rod.

The uniform applied magnetic field is defined as:

$$\mathbf{H}_{ext}(s) = H_{ext} \hat{\mathbf{H}}_{ext}(s),$$

where

$$\hat{\mathbf{H}}_{ext} = \cos \beta \mathbf{e}_3 + \sin \beta \mathbf{e}_1.$$

The angle made by the external magnetic field with respect to $\mathbf{e}_3$ is represented by β .

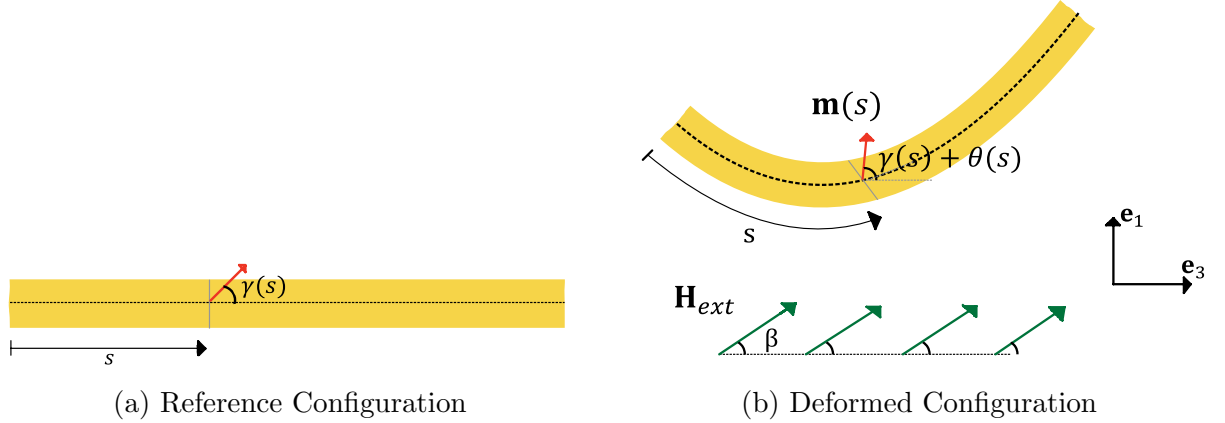


Figure 2.3: Kinematic description of a magnetoelastic rod. (a) Reference configuration showing the magnetization angle $\gamma(s)$, attached to the cross section. (b) Deformed configuration under an external magnetic field $\mathbf{H}_{ext}$, with local tangent rotation $\theta(s)$ generating the magnetic torque.

The magnetic torque may be expressed in terms of the local orientation as

$$\boldsymbol{\tau}(s) = A \mathbf{m}(s) \times \mathbf{H}_{ext} = C \tau(s) \mathbf{e}_2,$$

where $C = AM_{sat}H_{ext}$ and

$$\tau(s) = \sin(\beta - \theta(s) - \gamma(s)).$$

Thus, the dependence of magnetic torque on the angular variable $\gamma(s)$ is made explicit. This term provides a mechanism to control the deformation of the rod and plays a key role in eliminating lateral forces. From hereon, $\tau(s)$ is referred to as magnetic torque.

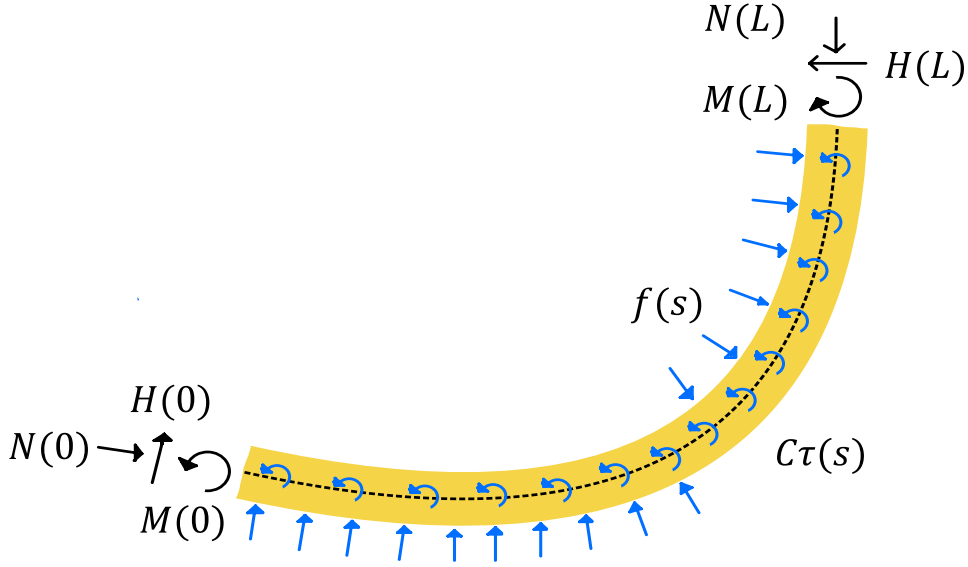


Figure 2.4: Free-body diagram of the magnetoelastic rod constrained within a channel, showing the distributed contact force $f(s)$, magnetic actuation torque $\tau(s)$, and the boundary forces and moment at the rod ends.

2.8 Continuous and Discontinuous Curvature

The channel curvature $k(s)$ may exhibit different levels of regularity. In this work, two cases are considered:

- **Continuous curvature:** The function $k(s)$ is smooth and differentiable along the length of the rod.
- **Discontinuous curvature:** The function $k(s)$ exhibits jump discontinuities at certain points. These discontinuities lead to jumps in internal force and moment components.

The formulation developed in this chapter serves as the basis for the detailed analysis of these two cases in Chapters 3 and 4.

Chapter 3

Rod in a Channel with Continuous Curvature

This chapter develops the formulation for eliminating lateral contact forces in a magnetoelastic rod constrained within a prescribed channel of continuous curvature.

3.1 Elastic Rod in a Channel

3.1.1 Equilibrium Equations and Boundary Conditions

Using the Cosserat rod formulation introduced in Chapter 2, we now specialize the governing equations to a planar, inextensible, and unshearable elastic rod constrained to follow a prescribed channel.

The equilibrium configuration is obtained by minimizing the total potential energy of the rod subject to the imposed constraints.

Variational Formulation

The total potential energy of the system is given by

$$\Pi = \int_0^L \frac{1}{2} EI \theta_s^2(s) \, ds - \mathbf{F}_{ext} \cdot \mathbf{r}(0), \quad (3.1)$$

subject to the constraints:

$$\begin{aligned} \nu(s) &= 1, \\ \eta(s) &= 0, \\ \phi_\Gamma(\mathbf{r}) &= 0. \end{aligned}$$

The augmented Lagrangian, incorporating the constraints above, is written as

$$\mathcal{L} = \int_0^L \left\{ \underbrace{\frac{1}{2}EI\theta_s^2}_{\text{Bending Energy}} + \underbrace{N\{\nu - 1\}}_{\text{Inextensibility}} + \underbrace{H\eta}_{\text{Unshearability}} + \underbrace{f\phi_\Gamma(\mathbf{r})}_{\text{Channel Constraint}} \right\} ds - \underbrace{\mathbf{F}_{ext} \cdot \mathbf{r}(0)}_{\text{Work by External Force}}. \quad (3.2)$$

where N, H and f are Lagrange parameters enforcing the constraints. Physically, they represent the axial force, shear force, and lateral force, respectively. Taking the first variation (see Appendix chapter A) yields the equilibrium equations and the boundary conditions.

Equilibrium Equations

$$\mathbf{n}_s - f\nabla\phi_\Gamma(\mathbf{r}) = \mathbf{0}, \quad (3.3)$$

$$EI\theta_{ss}\mathbf{d}_2 + \mathbf{r}_s \times \mathbf{n} = \mathbf{0}. \quad (3.4)$$

From $\nu(s) = 1$ and $\eta(s) = 0$, we get

$$\mathbf{r}_s(s) = \mathbf{d}_3(s) \quad \text{and} \quad \mathbf{r}(s) \times \mathbf{n}(s) = H(s)\mathbf{d}_2.$$

From $\phi_\Gamma(\mathbf{r}) = 0$, we get

$$\mathbf{r}(s) = \mathbf{\Gamma}(s + s_0) \quad \text{and} \quad \theta(s) = \Theta(s_0 + s),$$

where s_0 is the curvilinear coordinate of the left end of the rod inside the channel.

Let $k_c(s_0 + s) = \Theta_s(s_0 + s)$ and $\kappa(s) = \theta_s(s)$. Then

$$\kappa(s) = k_c(s_0 + s).$$

Boundary Conditions

From the first variation, we get

$$-\mathbf{F}_{ext} \cdot \delta\mathbf{r}(0) + \mathbf{n} \cdot \delta\mathbf{r} \Big|_0^L + EI\theta_s\delta\theta \Big|_0^L = 0. \quad (3.5)$$

Since external force is specified, $\delta\mathbf{r}(0) \neq 0$. As the centerline of the rod coincides with that of the channel, $\delta\mathbf{r}$ and $\delta\theta$ are not independent at the boundaries. Thus, we get

$$\delta\mathbf{r}(0) = \mathbf{\Gamma}_s(s_0)\delta s_0, \quad \delta\mathbf{r}(L) = \mathbf{\Gamma}_s(s_L)\delta s_L,$$

$$\delta\theta(0) = \Theta_s(s_0) \delta s_0, \quad \text{and} \quad \delta\theta(L) = \Theta_s(s_L) \delta s_L.$$

Using the above expressions in eq. (3.5), we get the boundary conditions as

$$N(0) + \mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) + EI\kappa^2(0) = 0 \quad (3.6)$$

$$N(L) + EI\kappa^2(L) = 0. \quad (3.7)$$

3.1.2 Expression for Lateral Force and External Force

Projecting eq. (3.3) and eq. (3.4) onto the director frame, we get

$$\begin{aligned} N_s - \kappa H &= 0, \\ \kappa N + H_s - f &= 0 \quad \text{and} \\ EI\kappa_s + H &= 0. \end{aligned}$$

Thus, we have

$$f = \kappa N + H_s,$$

where

$$N = -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI\kappa^2(0) + \int_0^s \kappa(s')H(s') \, ds' \quad (3.8)$$

and

$$H = -EI\kappa_s. \quad (3.9)$$

Putting the above expressions into f , we get

$$f(s) = -\frac{EI}{2}\kappa(s)\kappa^2(L) - \frac{EI}{2}\kappa^3(s) - EI\kappa_{ss}(s).$$

Using the expression for axial force, eq. (3.8) along with the boundary conditions, eq. (3.6) and eq. (3.7), we get the expression of external force as

$$\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) = \frac{EI}{2}\kappa^2(L) - \frac{EI}{2}\kappa^2(0).$$

3.2 Magnetoelastic Rod in a Channel

3.2.1 Equilibrium Equations and Boundary Conditions

The total potential energy now includes magnetic energy:

$$\Pi = \int_0^L \left[\frac{1}{2} EI \theta_s^2 - A \mathbf{m} \cdot \mathbf{H}_{ext} \right] ds - \mathbf{F}_{ext} \cdot \mathbf{r}(0). \quad (3.10)$$

The equilibrium equations become

$$\mathbf{n}_s - f \nabla \phi_\Gamma = 0, \quad (3.11)$$

$$EI \theta_{ss} \mathbf{d}_2 + \mathbf{r}_s \times \mathbf{n} + A \mathbf{m} \times \mathbf{H}_{ext} = 0. \quad (3.12)$$

The expressions for the boundary conditions remain as eq. (3.6) and eq. (3.7).

3.2.2 Expression for Lateral Force and External Force

Projecting eq. (3.11) and eq. (3.12) onto the director frame, we get

$$\begin{aligned} N_s - \kappa H &= 0, \\ \kappa N + H_s - f &= 0 \\ \text{and} \quad EI \kappa_s + H + C \tau &= 0. \end{aligned}$$

Thus, we have

$$f = \kappa N + H_s$$

where

$$N = -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI \kappa^2(0) + \int_0^s \kappa(s') H(s') ds' \quad (3.13)$$

and

$$H = -EI \kappa_s - C \tau. \quad (3.14)$$

Putting the above expressions into f , we get (see Appendix chapter B)

$$f(s) = -\frac{EI}{2} \kappa(s) \kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI \kappa_{ss}(s) + C \kappa(s) \int_s^L \kappa(s') \tau(s') ds' - C \tau_s(s). \quad (3.15)$$

Using eq. (3.6), eq. (3.7) and eq. (3.13), we get the expression of external force as

$$\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) = \frac{EI}{2} \kappa^2(L) - \frac{EI}{2} \kappa^2(0) - C \int_0^L \kappa(s') \tau(s') ds'. \quad (3.16)$$

3.3 Elimination of Lateral Force

3.3.1 Governing Differential Equation and Boundary Conditions

Let

$$\Lambda_\Gamma(s) = -\frac{EI}{2} \kappa(s) \kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI \kappa_{ss}(s).$$

Thus, eq. (3.15) simplifies to

$$f(s) = \Lambda_\Gamma(s) + C \kappa(s) \int_s^L \kappa(s') \tau(s') ds' - C \tau_s(s).$$

Setting $f(s) = 0$ gives

$$C \tau_s(s) = \Lambda_\Gamma(s) + C \kappa(s) \int_s^L \kappa(s') \tau(s') ds'. \quad (3.17)$$

Differentiating eq. (3.17) gives

$$C \tau_{ss}(s) = \Lambda_{\Gamma,s}(s) + C \kappa_s(s) \int_s^L \kappa(s') \tau(s') ds' - C \kappa^2(s) \tau(s).$$

Assuming $\kappa(s) \neq 0$, substituting the expression for $\int_s^L \kappa(s') \tau(s') ds'$ from eq. (3.17), we get the governing differential equation as:

$$\kappa(s) \tau_{ss}(s) - \kappa_s(s) \tau_s(s) + \kappa^3(s) \tau(s) - \psi_\Gamma(s) = 0, \quad (3.18)$$

where

$$\psi_\Gamma(s) = \frac{1}{C} \left(\Lambda_{\Gamma,s}(s) \kappa(s) - \kappa_s(s) \Lambda_\Gamma(s) \right).$$

Boundary condition from eq. (3.17):

$$\tau_s(L) = \frac{\Lambda_\Gamma(L)}{C} = -\frac{1}{C} \left(EI \kappa^3(L) + EI \kappa_{ss}(L) \right). \quad (3.19)$$

Another boundary condition is obtained by specifying magnetic torque at $s = 0$:

$$\tau(0) = \sin(\beta - \theta(0) - \gamma(0)) = \tau_0. \quad (3.20)$$

Since the magnetization angle at the boundary, $\gamma(0)$, can be prescribed independently, the boundary value τ_0 can be specified for a chosen β . Thus, τ_0 acts as a controllable parameter in the present formulation.

3.3.2 General Solution

We have

$$\psi_\Gamma(s) = \frac{1}{C} \left(\Lambda_{\Gamma,s}(s) \kappa(s) - \kappa_s(s) \Lambda_\Gamma(s) \right).$$

Simplifying the expression, we get

$$\psi_\Gamma(s) = -\frac{EI}{C} \left(\kappa \kappa_{sss} - \kappa_s \kappa_{ss} + \kappa^3 \kappa_s \right).$$

Thus, the differential equation simplifies to

$$\kappa \tau_{ss} - \kappa_s \tau_s + \kappa^3 \tau = -\frac{EI}{C} \left(\kappa \kappa_{sss} - \kappa_s \kappa_{ss} + \kappa^3 \kappa_s \right). \quad (3.21)$$

The general solution for eq. (3.21) is (see Appendix chapter C)

$$\tau(s) = \tau_h(s) + \tau_p(s),$$

where

$$\tau_h(s) = A_1 \sin[\theta(s) - \theta(0)] + B_1 \cos[\theta(s) - \theta(0)], \quad \text{and}$$

$$\tau_p(s) = -\frac{EI}{C} \int_0^s g(s') \sin[\theta(s) - \theta(s')] ds'.$$

Thus, the magnetic torque is given as

$$\tau(s) = A_1 \sin[\theta(s) - \theta(0)] + B_1 \cos[\theta(s) - \theta(0)] - \frac{EI}{C} \int_0^s g(s') \sin[\theta(s) - \theta(s')] ds', \quad (3.22)$$

where

$$g(s) = \frac{d}{ds} \left(\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} \right).$$

Now,

$$\tau_{p,s}(s) = \frac{EI}{C} \kappa(s) \int_0^s g(s') \cos [\theta(s) - \theta(s')] ds'.$$

Using eq. (3.19) and eq. (3.20), we get

$$A_1 = \tau_0 \tan [\theta(L) - \theta(0)] + \frac{\tau_s(L) - \tau_{p,s}(L)}{\kappa(L) \cos [\theta(L) - \theta(0)]} \quad \text{and} \quad (3.23)$$

$$B_1 = \tau_0. \quad (3.24)$$

The general solution is valid for channel curvatures $\kappa(s)$ which satisfy the constraint

$$|\tau(s)| \leq 1. \quad (3.25)$$

3.3.3 Remarks on Degenerate Cases

The derivation of the governing differential equation and the expressions for the integration constants assume that

$$\kappa(s) \neq 0, \quad \kappa(L) \neq 0, \quad \cos [\theta(L) - \theta(0)] \neq 0.$$

Special care is required when one or more of these conditions are violated.

Case 1: Straight Channel

If

$$\kappa(s) = 0,$$

the substitution used in the derivation of eq. (3.18) becomes singular. In this case, the channel corresponds to a straight segment, and the lateral force expression reduces to

$$f(s) = -C\tau_s(s).$$

Imposing the condition $f(s) = 0$ therefore gives $\tau_s(s) = 0$. Hence,

$$\tau(s) = \tau_0,$$

where τ_0 is constant along the straight segment.

Thus, straight portions of the channel correspond to a constant magnetic torque distribution.

Case 2: Degeneracy in Integration Constant

Using the general solution

$$\tau(s) = A_1 \sin [\theta(s) - \theta(0)] + \tau_0 \cos [\theta(s) - \theta(0)] + \tau_p(s),$$

the derivative is

$$\tau_s(s) = \kappa(s) \left[A_1 \cos [\theta(s) - \theta(0)] - \tau_0 \sin [\theta(s) - \theta(0)] \right] + \tau_{p,s}(s).$$

Evaluating the above expression at $s = L$ gives

$$\tau_s(L) = \kappa(L) \left[A_1 \cos [\theta(L) - \theta(0)] - \tau_0 \sin [\theta(L) - \theta(0)] \right] + \tau_{p,s}(L). \quad (3.26)$$

If either

$$\kappa(L) = 0$$

or

$$\cos [\theta(L) - \theta(0)] = 0,$$

the dependence on the coefficient A_1 disappears from eq. (3.26). Consequently, the boundary condition at $s = L$ no longer uniquely determines the homogeneous component of the solution. Instead, the boundary condition reduces to a compatibility condition.

For the case

$$\kappa(L) = 0,$$

eq. (3.26) reduces to

$$\tau_s(L) = \tau_{p,s}(L).$$

Similarly, when

$$\theta(L) - \theta(0) = (2n + 1)\frac{\pi}{2}, n \in \mathbf{Z}$$

we obtain

$$\tau_s(L) - \tau_{p,s}(L) = -\kappa(L)\tau_0 \sin [\theta(L) - \theta(0)].$$

Thus, in both degenerate cases, the coefficient A_1 remains undetermined by the boundary conditions.

However, the magnetic torque must satisfy eq. (3.25). Substituting the general solution

yields the admissibility condition

$$\left| A_1 \sin [\theta(s) - \theta(0)] + \tau_0 \cos [\theta(s) - \theta(0)] + \tau_p(s) \right| \leq 1.$$

Therefore, although A_1 is not uniquely determined by the boundary conditions, it remains restricted to values satisfying the above inequality throughout the domain.

Hence, the degenerate cases lead to non-uniqueness in the homogeneous component of the solution while still preserving bounded admissible solution families.

3.3.4 Recovering Magnetization Angle from Magnetic Torque

Once the magnetic torque distribution $\tau(s)$ has been determined, the corresponding magnetization angle $\gamma(s)$ can be recovered from the relation

$$\tau(s) = \sin (\beta - \theta(s) - \gamma(s)).$$

Inverting the above relation yields

$$\beta - \theta(s) - \gamma(s) = \sin^{-1}[\tau(s)].$$

Hence,

$$\gamma(s) = \beta - \theta(s) - \sin^{-1}[\tau(s)].$$

However, since the sine function is not one-to-one, an additional admissible branch exists:

$$\beta - \theta(s) - \gamma(s) = \pi - \sin^{-1}[\tau(s)].$$

This gives the alternative solution

$$\gamma(s) = \beta - \theta(s) - (\pi - \sin^{-1}[\tau(s)]).$$

Thus, the reconstruction of $\gamma(s)$ from $\tau(s)$ is generally non-unique, and either branch may be chosen since both satisfy the governing relation.

3.4 Solutions for Canonical Curvatures

In this section, analytical solutions for the magnetic torque $\tau(s)$ are obtained for canonical channel geometries. These examples illustrate the general framework developed in the preceding sections.

3.4.1 Circular Channel

For a circular channel, the curvature is constant:

$$\kappa(s) = K.$$

In this case, $\tau_p(s) = 0$, and the governing equation reduces to a homogeneous form. The general solution is

$$\tau(s) = A_1 \sin Ks + B_1 \cos Ks.$$

Using eq. (3.23) and eq. (3.24), the constants are obtained as

$$A_1 = \tau_0 \tan(KL) - \frac{EIK^2}{C \cos(KL)} \quad \text{and} \quad B_1 = \tau_0.$$

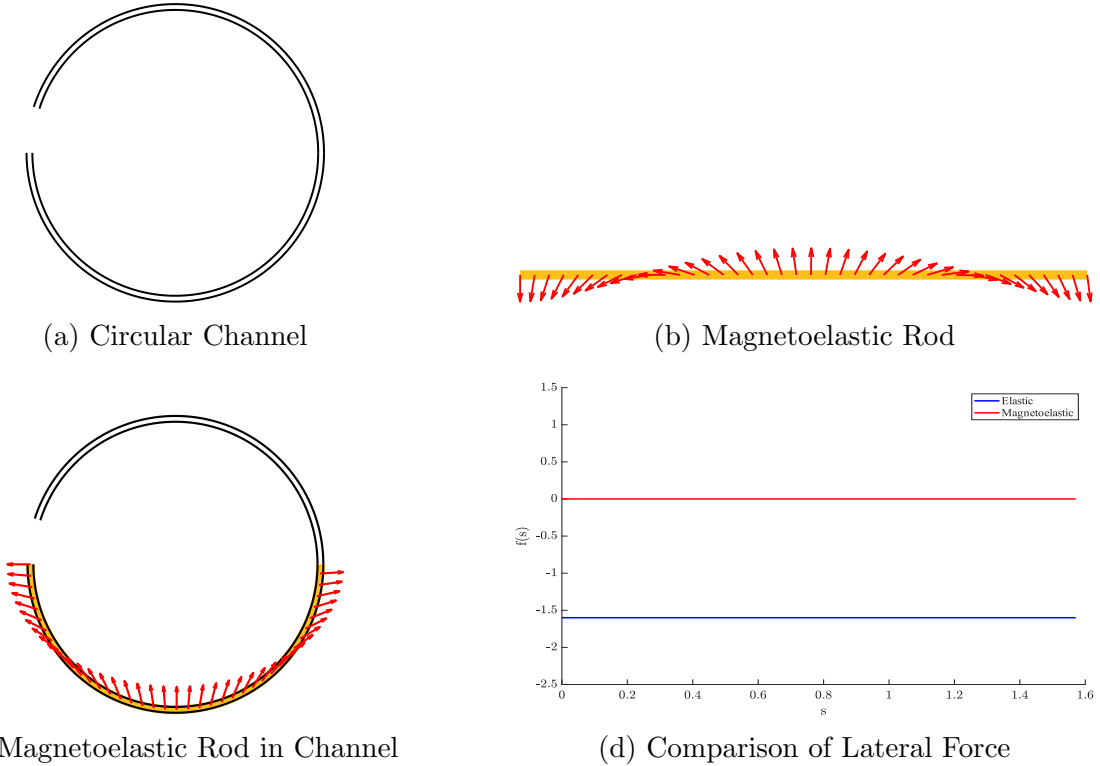


Figure 3.1: Comparison of elastic and magnetoelastic behavior in a circular channel. (a) Prescribed circular channel geometry, with $K = 2$ and $\theta(0) = -90^\circ$. (b) Magnetoelastic rod of length $L = \pi$ with the corresponding embedded magnetization distribution having $\gamma(0) = -90^\circ$ obtained for $H_{ext} = 0.45$ and $\beta = 90^\circ$. (c) Equilibrium configuration of the magnetoelastic rod within the channel. (d) Comparison of the lateral force, showing elimination of the lateral force in the magnetoelastic case.

3.4.2 Spiral Channel

For a spiral channel, the curvature varies linearly:

$$\kappa(s) = Ks.$$

The general solution consists of a homogeneous and a particular part:

$$\tau(s) = A_1 \sin \frac{Ks^2}{2} + B_1 \cos \frac{Ks^2}{2} - \frac{EIK}{C} \left(1 - \cos \frac{Ks^2}{2} \right).$$

Applying the boundary conditions eq. (3.22) and eq. (3.23) yields the following result

$$A_1 = \tau_0 \tan \left(\frac{KL^2}{2} \right) - \frac{EIK^2L^2}{C \cos \left(\frac{KL^2}{2} \right)} - \frac{EIK}{C} \tan \left(\frac{KL^2}{2} \right) \quad \text{and} \quad B_1 = \tau_0.$$

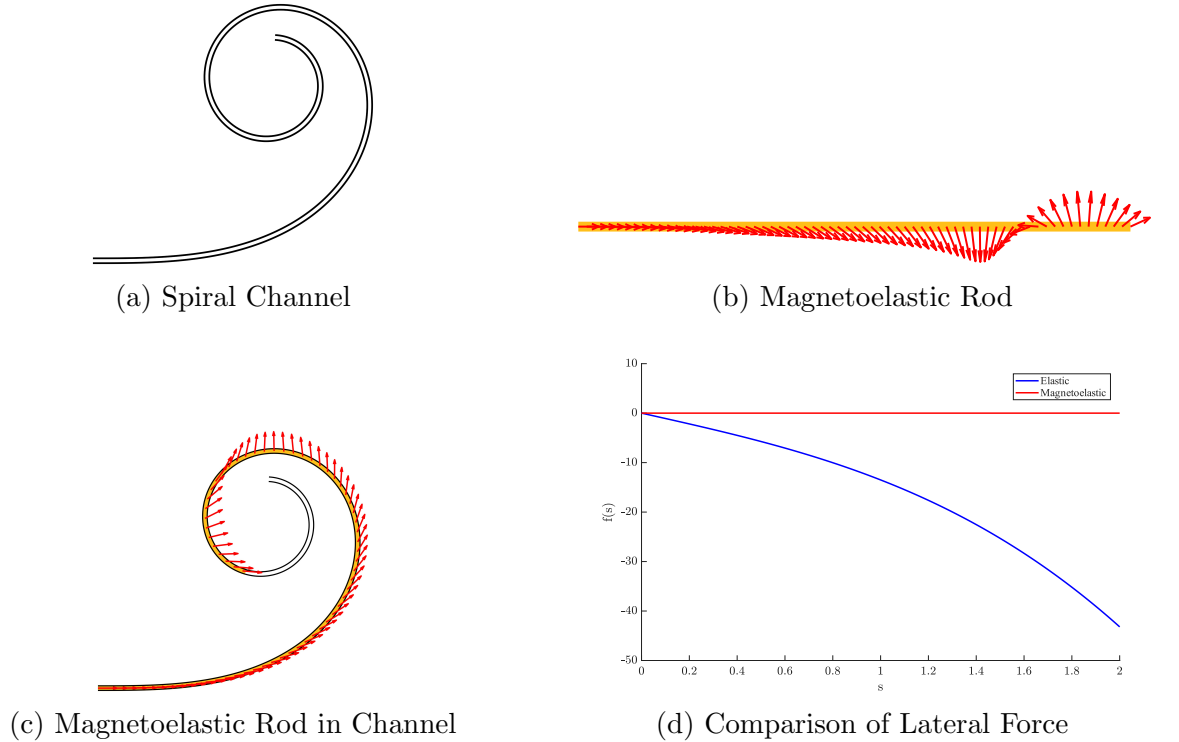


Figure 3.2: Comparison of elastic and magnetoelastic behavior in a spiral channel. (a) Prescribed spiral channel geometry, with $K = 3$ and $\theta(0) = 0^\circ$. (b) Magnetoelastic rod of length $L = 2$ with the corresponding embedded magnetization distribution having $\gamma(0) = 0^\circ$ obtained for $H_{ext} = 0.8$ and $\beta = 45^\circ$. (c) Equilibrium configuration of the magnetoelastic rod within the channel. (d) Comparison of the lateral force, showing elimination of the lateral force in the magnetoelastic case.

3.4.3 Sinusoidal Channel

For a sinusoidal channel, the curvature is given by

$$\kappa(s) = X \sin\left(\frac{s}{Y}\right).$$

For small values of X , the channel is close to a straight segment, while as X grows, it becomes wavier. The parameter Y dictates how many turns the channel has per unit length [1]. Thus, $\theta(s)$ is obtained as

$$\theta(s) = -XY \cos\left(\frac{s}{Y}\right) + XY + \theta(0).$$

The general solution for $\tau(s)$ is

$$\tau(s) = A_1 \sin \theta(s) + B_1 \cos \theta(s) + \tau_p(s).$$

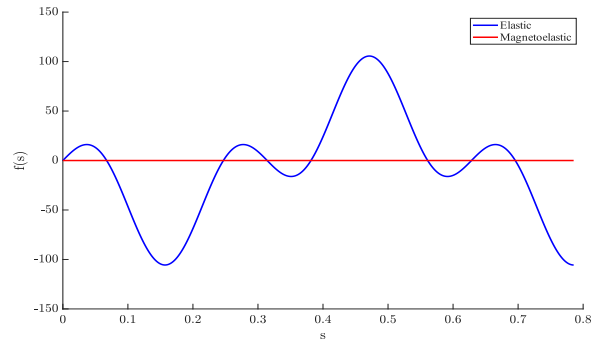
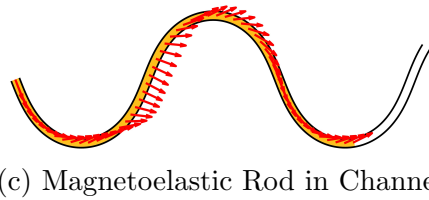
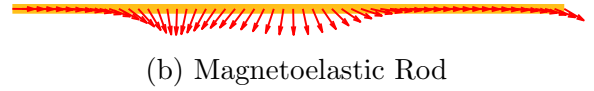
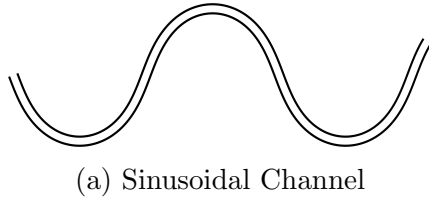


Figure 3.3: Comparison of elastic and magnetoelastic behavior in a sinusoidal channel. (a) Prescribed sinusoidal channel geometry, with $X = 12$, $Y = 0.1$ and $\theta(0) = -XY$. (b) Magnetoelastic rod of length $L = 2\pi Y$ with the corresponding embedded magnetization distribution having $\gamma(0) = 0^\circ$ obtained for $H_{ext} = 0.8$ and $\beta = -90^\circ$. (c) Equilibrium configuration of the magnetoelastic rod within the channel. (d) Comparison of the lateral force, showing elimination of the lateral force in the magnetoelastic case.

The particular solution is obtained as

$$\begin{aligned} \tau_p(s) = & -\frac{EI}{CY^2} \left(-XY \cos\left(\frac{s}{Y}\right) + XY + \sin \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] \right) \\ & - \frac{EI\theta(0)}{CY^2} \left(1 - \cos \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] \right). \end{aligned}$$

The expressions for A_1 and B_1 are determined using eq. (3.23) and eq. (3.24) as in the previous cases.

The expression for the magnetization angle $\gamma(s)$ can be recovered from the expressions of $\tau(s)$ for each channel.

3.5 Rod Insertion

The formulation developed in section 3.3 determines the magnetization angle $\gamma(s)$ required for a rod of a given length to conform to a prescribed section of channel. However, during practical rod insertion, the length of the rod inside the channel changes continuously. Therefore, the solution obtained using eq. (3.22) gives a different magnetization distribution for each length. In a practical scenario, it is difficult to have an active control over the magnetization angle. Hence, such a strategy is not feasible.

Therefore, for rod insertion, it is desirable to identify a class of solutions for which the embedded magnetization remains independent of the inserted length. One such configuration is that of a constant magnetization angle,

$$\gamma(s) = \gamma_0,$$

which corresponds to a uniformly magnetized rod relative to the fixed reference frame. Thus, constant- γ solutions provide simplified actuation strategies for rod insertion.

For the rod insertion problems considered here, the arc-length coordinate is chosen such that the left end of the rod inside the channel coincides with the channel entry. Hence,

$$s_0 = 0,$$

and the inserted rod occupies the interval $[0, L]$.

Let

$$\tau(s) = \sin(\beta - \theta(s) - \gamma_0).$$

Substituting this expression into the governing differential eq. (3.21), we obtain

$$\kappa\kappa_{sss} - \kappa_s\kappa_{ss} + \kappa^3\kappa_s = 0, \quad (3.27)$$

which is a function of channel geometry alone. Clearly,

$$\kappa(s) = 0 \quad \text{and} \quad \kappa(s) = K$$

satisfy (3.27). However,

$$\kappa(s) = Ks \quad \text{and} \quad \kappa(s) = X \sin\left(\frac{s}{Y}\right)$$

do not satisfy the equation. Thus, constant- γ rod insertion is not admissible in spiral or sinusoidal channels.

3.5.1 Channel Curvature for Rod Insertion

Assuming $\kappa(s) \neq 0$, dividing eq. (3.27) by κ^2 gives

$$\frac{d}{ds} \left(\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} \right) = 0, \quad (3.28)$$

which integrates to

$$\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} = \omega_1. \quad (3.29)$$

A second integration yields

$$\kappa_s^2 = -\frac{\kappa^4}{4} + \omega_1\kappa^2 + \omega_2. \quad (3.30)$$

Introducing the roots of the quartic and defining

$$a^2 = 2\omega_1 + 2\sqrt{\omega_1^2 + \omega_2}, \quad b^2 = 2\omega_1 - 2\sqrt{\omega_1^2 + \omega_2},$$

the equation becomes

$$\kappa_s^2 = \frac{1}{4}(a^2 - \kappa^2)(\kappa^2 - b^2). \quad (3.31)$$

This is an elliptic differential equation, whose solution is (see Appendix chapter F)

$$\kappa(s) = a \operatorname{cn} \left(\frac{a}{2\sqrt{m}}(s - \omega_3), m \right), \quad (3.32)$$

where $\text{cn}(u, m)$ is the *Jacobi elliptic cosine*, defined via

$$\text{cn}(u, m) = \cos(\phi), \quad u = \int_0^\phi \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}}, \quad \text{and } m = \frac{a^2 - b^2}{a^2}.$$

An analytical solution is obtained when $m = 1$, corresponding to $\omega_2 = 0$, for which

$$\text{cn}(u, 1) = \text{sech}(u),$$

giving

$$\kappa(s) = 2\lambda \text{sech}[\lambda(s - \omega_3)], \quad (3.33)$$

where $\lambda = \sqrt{\omega_1}$. Thus, constant-magnetization-angle rod insertion is compatible with a family of channel geometries parameterized by elliptic functions, along with straight and circular channels.

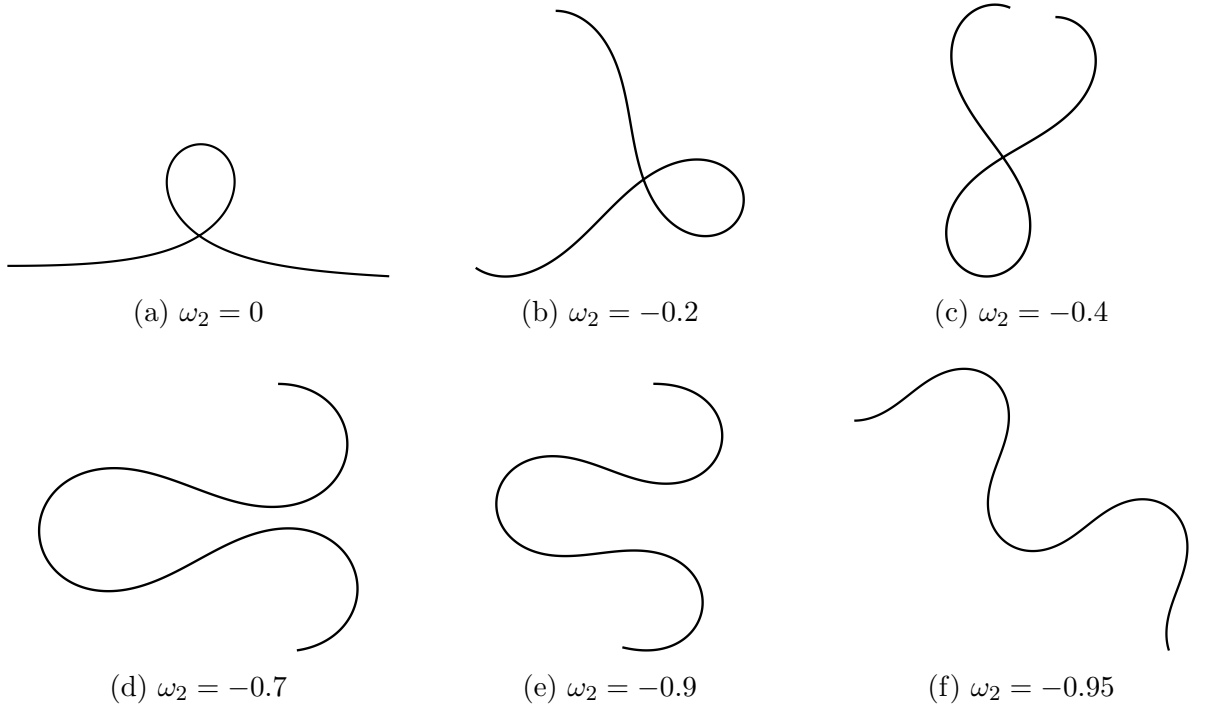


Figure 3.4: Admissible channel geometries for rod insertion with constant magnetization angle, generated from curvature functions satisfying eq. (3.32). Figures (a)–(f) are plotted with $\omega_1 = 1$, $\omega_3 = 5$ and $L = 10$. Figure (a) shows the plot corresponding analytical solution.

3.5.2 Expression for Constant Magnetization Angle

Now,

$$\begin{aligned}\tau(s) &= \sin(\beta - \theta(s) - \gamma_0) \\ &= \sin(\beta - \theta(0) - \gamma_0) \cos[\theta(s) - \theta(0)] - \cos(\beta - \theta(0) - \gamma_0) \sin[\theta(s) - \theta(0)].\end{aligned}$$

Since $g(s) = 0$, $\tau_p(s) = 0$. Thus, from eq. (3.22), we have

$$\tau(s) = A_1 \sin[\theta(s) - \theta(0)] + B_1 \cos[\theta(s) - \theta(0)].$$

Comparing the expressions using eq. (3.23) and eq. (3.24), we get

$$\cos(\beta - \theta(L) - \gamma_0) = \cos \left[\pm \cos^{-1} \left(\frac{\tau_s(L)}{\kappa(L)} \right) \right] = \cos \left[\pm \cos^{-1} \left(\frac{EI \kappa^2(L)}{C} + \frac{EI \kappa_{ss}(L)}{C \kappa(L)} \right) \right].$$

We know,

$$\frac{\kappa^2(L)}{2} + \frac{\kappa_{ss}(L)}{\kappa(L)} = \omega_1.$$

Thus,

$$\gamma_0 = \beta - \theta(L) \pm \cos^{-1} \left[\frac{EI}{C} \left(\frac{\kappa^2(L)}{2} + \omega_1 \right) \right]. \quad (3.34)$$

3.5.3 Quasi-Static Rod Insertion Using Constant Magnetization

The expression derived in eq. (3.34) provides a practical strategy for rod insertion using a uniformly magnetized rod.

Consider a rod with a prescribed constant embedded magnetization angle,

$$\gamma(s) = \gamma_0.$$

A desired final inserted configuration of length L_f is first selected. For this final equilibrium configuration, the corresponding external magnetic field orientation is obtained from eq. (3.34) as

$$\beta_f = \gamma_0 + \theta(L_f) \mp \cos^{-1} \left[\frac{EI}{C} \left(\frac{\kappa^2(L_f)}{2} + \omega_1 \right) \right].$$

Now consider an intermediate insertion state corresponding to an inserted length $L <$

L_f . Since the embedded magnetization remains unchanged, the same constant value γ_0 must continue to satisfy the equilibrium condition. Therefore,

$$\gamma_0 = \beta(L) - \theta(L) \pm \cos^{-1} \left[\frac{EI}{C} \left(\frac{\kappa^2(L)}{2} + \omega_1 \right) \right],$$

from which the required magnetic field orientation $\beta(L)$ may be determined.

Simultaneously, the holding force required to maintain equilibrium at that insertion state is obtained from eq. (3.14).

Thus, rod insertion may be achieved quasi-statically by maintaining a fixed embedded magnetization profile while continuously adjusting the external magnetic field orientation and the applied holding force in accordance with the instantaneous inserted length.

This provides a physically realizable actuation strategy for controlled insertion of magnetoelastic rods into admissible channel geometries.

3.5.4 Rod Insertion in a Circular Channel

For a circular channel, $\theta(L) = \theta(0) + KL$, $\kappa(L) = K$ and $\omega_1 = \frac{K^2}{2}$. Thus, using eq. (3.34), we get

$$\gamma_0 = \beta - \theta(0) - KL \pm \cos^{-1} \left(\frac{EIK^2}{C} \right).$$

Using the strategy discussed in section 3.5.3, we can have rod insertion as shown in fig. 3.5.

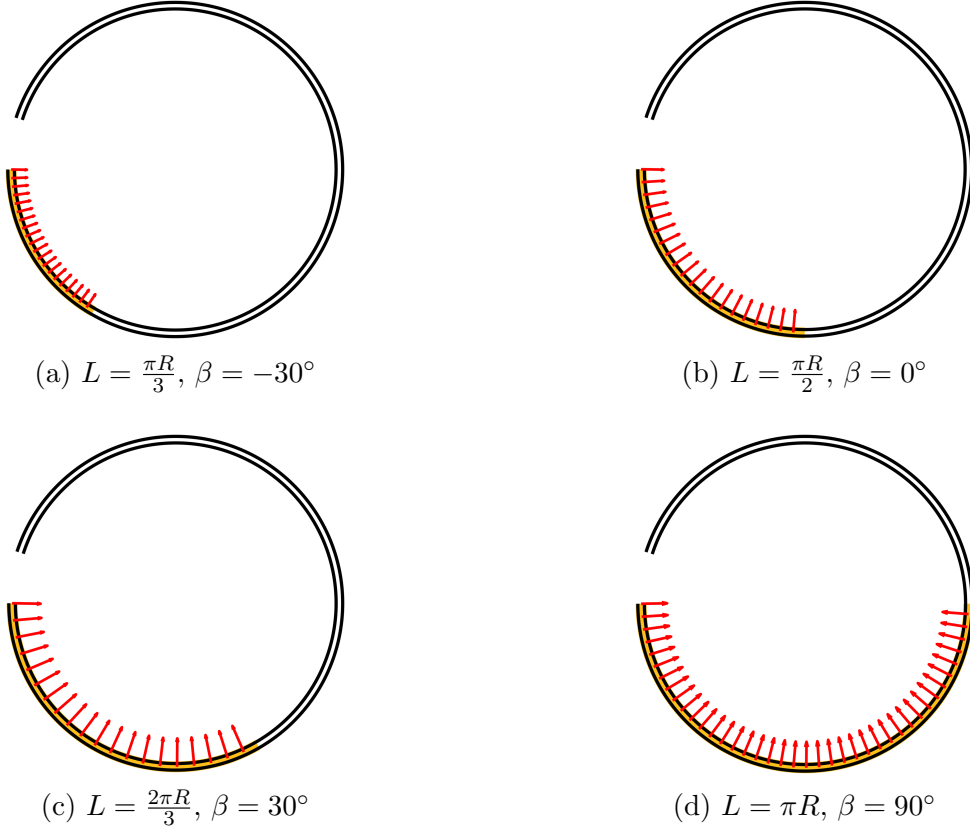


Figure 3.5: Quasi-static insertion of a uniformly magnetized rod into a circular channel of radius $R = 0.5$. Figures (a)–(d) show successive equilibrium configurations corresponding to increasing inserted length L , with the external magnetic field orientation β adjusted to maintain the constant magnetization angle $\gamma_0 \approx 90^\circ$ at $H_{ext} = 0.45$.

Chapter 4

Rod in a Channel with Discontinuous Curvature

4.1 Elastic Rod in a Channel

4.1.1 Equilibrium Equations and Boundary Conditions

The governing equilibrium equations and boundary conditions remain identical to those derived for the continuous curvature case for an elastic rod in section 3.1.1. For completeness, they are summarized below.

Equilibrium Equations

$$\begin{aligned} N_s - \kappa H &= 0 \\ \kappa N + H_s - f &= 0 \\ EI\kappa_s + H &= 0 \end{aligned}$$

Boundary Conditions

$$\begin{aligned} N(0) + \mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) + EI\kappa^2(0) &= 0 \\ N(L) + EI\kappa^2(L) &= 0. \end{aligned}$$

The key distinction in the present case arises from the discontinuity in curvature, which leads to jump conditions in the internal force and moment components at the interface.

4.1.2 Jump Conditions at Discontinuity

Let the curvature $\kappa(s)$ be defined as:

$$\kappa(s) = \begin{cases} \kappa_1(s) & s < l \\ \kappa_2(s) & l < s, \end{cases}$$

so that there is a jump discontinuity at $s = l$ in curvature. While the curvature $\kappa(s)$ is allowed to exhibit jump discontinuities at the interface, the centerline and $\theta(s)$ remain continuous. As a result, the rod centerline remains geometrically continuous across $s = l$, i.e., the configuration is C^0 continuous but not C^1 . Physically, this implies that while the rod does not exhibit any geometric break or kink, the rate of bending changes abruptly at the interface.

This discontinuity in curvature leads to corresponding discontinuities in the internal force and moment, which manifest as concentrated forces and concentrated moment at $s = l$.

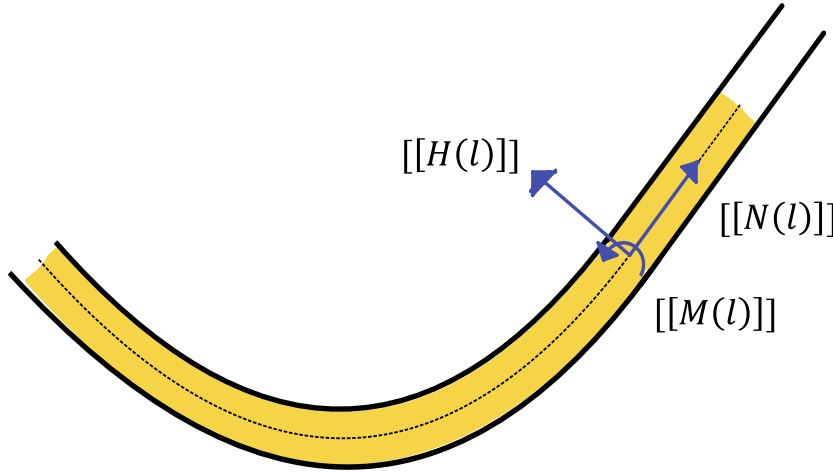


Figure 4.1: Interface loading associated with a curvature discontinuity in the prescribed channel geometry. The jumps in axial force, shear force, and bending moment are denoted by $\llbracket N(l) \rrbracket$, $\llbracket H(l) \rrbracket$, and $\llbracket M(l) \rrbracket$, respectively.

Using the equilibrium equations yields a jump in internal forces and moments across the discontinuity (see Appendix chapter D):

$$\llbracket N(l) \rrbracket = -\frac{EI}{2} \llbracket \kappa^2(l) \rrbracket,$$

$$\llbracket H(l) \rrbracket = -EI \llbracket \kappa_s(l) \rrbracket, \quad \text{and}$$

$$\llbracket M(l) \rrbracket = EI \llbracket \kappa(l) \rrbracket.$$

These correspond to:

- point force along $\mathbf{d}_3$,
- point force along $\mathbf{d}_1$,
- concentrated moment along $\mathbf{d}_2$.

The jump in axial force is due to the configurational force, discussed in [2]

4.1.3 Expression for Lateral Force and External Force

Using the equilibrium equations, we get the expression of the lateral force as (see Appendix chapter E)

$$f(s) = -\frac{EI}{2} \kappa(s) \kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI \kappa_{ss}(s).$$

The expression for the external force is

$$\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) = \frac{EI}{2} \kappa^2(L) - \frac{EI}{2} \kappa^2(0).$$

It is noted that the expressions for the lateral force $f(s)$ and $\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0)$ retain the same formal structure as in the continuous curvature case. However, their interpretation differs due to the presence of curvature discontinuities.

To make the effect of curvature discontinuity explicit, the curvature can be represented using a Heaviside function as

$$\kappa(s) = \kappa_1(s) + [\kappa_2(s) - \kappa_1(s)] H(s - l), \quad (4.1)$$

where $H(s - l)$ is the Heaviside function defined as

$$H(s - l) = \begin{cases} 0, & s < l, \\ 1, & s \geq l. \end{cases}$$

First, in the evaluation of the boundary term, the curvature must be interpreted piecewise. In particular, the curvature values at the boundaries correspond to those of the respective

segments, i.e.,

$$\kappa(0) = \kappa_1(0), \quad \kappa(L) = \kappa_2(L).$$

Thus, the boundary contribution depends on the curvature of the segment to which each boundary point belongs.

Second, the expression for the lateral force involves the second derivative of curvature, $\kappa_{ss}(s)$. Differentiating eq. (4.1), we obtain

$$\kappa_s(s) = \kappa_{1,s}(s) + [\kappa_{2,s}(s) - \kappa_{1,s}(s)]H(s-l) + \llbracket \kappa_s(l) \rrbracket \delta(s-l),$$

and

$$\kappa_{ss}(s) = \kappa_{1,ss}(s) + [\kappa_{2,ss}(s) - \kappa_{1,ss}(s)]H(s-l) + \llbracket \kappa_s(l) \rrbracket \delta(s-l) + \llbracket \kappa(l) \rrbracket \delta_s(s-l).$$

Thus, $\kappa_s(s)$ and $\kappa_{ss}(s)$ contain singular terms involving Dirac delta function and its derivative. These terms represent concentrated mechanical effects at the interface $s = l$.

In particular:

- the $\delta(s-l)$ term corresponds to a point force in the $\mathbf{d}_1$ direction:

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s') \, ds' = -EI \llbracket \kappa_s(l) \rrbracket.$$

- the $\delta_s(s-l)$ term corresponds to a concentrated moment about the $\mathbf{d}_2$ direction:

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s')(s' - l) \, ds' = EI \llbracket \kappa(l) \rrbracket.$$

These distributional contributions are consistent with the jump conditions derived earlier.

4.2 Magnetoelastic Rod in a Channel

4.2.1 Equilibrium Equations and Boundary Conditions

The governing equations and the boundary conditions remain identical to those derived for the continuous curvature case for an elastic rod in section 3.2.1. For completeness, they are summarized below.

Equilibrium Equations

$$\begin{cases} N_s - \kappa H = 0 \\ \kappa N + H_s - f = 0 \\ EI\kappa_s + H + C\tau = 0 \end{cases}$$

Boundary Conditions

$$\begin{cases} N(0) + \mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) + EI\kappa^2(0) = 0 \\ N(L) + EI\kappa^2(L) = 0. \end{cases}$$

The key distinction in the present case arises from the discontinuity in curvature, which leads to jump conditions in the internal force and moment components at the interface.

4.2.2 Jump Conditions at Discontinuity

Using the equilibrium equations yields a jump in internal forces and moments across the discontinuity (see Appendix chapter D):

$$\llbracket N(l) \rrbracket = -\frac{EI}{2} \llbracket \kappa^2(l) \rrbracket,$$

$$\llbracket H(l) \rrbracket = -EI \llbracket \kappa_s(l) \rrbracket - C \llbracket \tau(l) \rrbracket \quad \text{and}$$

$$\llbracket M(l) \rrbracket = EI \llbracket \kappa(l) \rrbracket.$$

These correspond to:

- point force along $\mathbf{d}_3$,
- point force along $\mathbf{d}_1$ and,
- concentrated moment along $\mathbf{d}_2$, respectively.

The magnetic torque $\tau(s)$ is assumed to have at most a finite admissible jump.

4.2.3 Expression for Lateral Force and External Force

Using the equilibrium equations, we get the expression of the lateral force as

$$f(s) = -\frac{EI}{2} \kappa(s) \kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI \kappa_{ss}(s) + C \kappa(s) \int_s^L \kappa(s') \tau(s') ds' - C \tau_s(s).$$

The expression for external force is

$$\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) = \frac{EI}{2} \kappa^2(L) - \frac{EI}{2} \kappa^2(0) - C \int_0^L \kappa(s') \tau(s') ds'.$$

As in the previous section, the expressions for the lateral force $f(s)$ and external force $\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0)$ retain the same formal structure as in the continuous curvature case. However, their interpretation differs due to the presence of curvature and magnetic torque discontinuities.

In particular:

- the $\delta(s - l)$ term corresponds to a point force in the $\mathbf{d}_1$ direction:

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s') ds' = -EI \llbracket \kappa_s(l) \rrbracket - C \llbracket \tau(l) \rrbracket.$$

- the $\delta_s(s - l)$ term corresponds to a concentrated moment about the $\mathbf{d}_2$ direction:

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s')(s' - l) ds' = EI \llbracket \kappa(l) \rrbracket.$$

These distributional contributions are consistent with the jump conditions derived earlier.

4.3 Elimination of Lateral Force

4.3.1 Governing Differential Equation

We have

$$f(s) = \Lambda_\Gamma(s) + C \kappa(s) \int_s^L \kappa(s') \tau(s') ds' - C \tau_s(s).$$

Setting $f(s) = 0$, we get

$$C \tau_s(s) = \Lambda_\Gamma(s) + \kappa(s) C \int_s^L \kappa(s') \tau(s') ds'. \quad (4.2)$$

Differentiating eq. (4.2).

$$C \tau_{ss}(s) = \Lambda_{\Gamma,s}(s) + C \kappa_s(s) \int_s^L \kappa(s') \tau(s') ds' - C \kappa^2(s) \tau(s).$$

Assuming $\kappa(s) \neq 0$, substituting the expression for the integral from eq. (4.2), we get the governing differential equation as:

$$\kappa(s)\tau_{ss}(s) - \kappa_s(s)\tau_s(s) + \kappa^3(s)\tau(s) - \psi_\Gamma(s) = 0, \quad (4.3)$$

where

$$\psi_\Gamma(s) = \frac{1}{C} \left(\Lambda_{\Gamma,s}(s)\kappa(s) - \kappa_s(s)\Lambda_\Gamma(s) \right).$$

The equation eq. (4.3) is solved separately for each channel segment.

4.3.2 Boundary and Interface Conditions

To solve eq. (4.3) in regions $(0, l)$ and (l, L) , we require two boundary conditions each. Setting $s = L$ in eq. (4.2), we get

$$\tau_s(L) = \frac{\Lambda_\Gamma(L)}{C}.$$

By prescribing $\gamma(0)$, we get another boundary condition:

$$\tau(0) = \tau_0.$$

From making the point load along $\mathbf{d}_1(l)$ to be 0 (or) the jump in shear force $\llbracket H(l) \rrbracket = 0$, we get

$$\llbracket \tau(l) \rrbracket = -\frac{EI}{C} \llbracket \kappa_s(l) \rrbracket.$$

Substituting $s = l + h$ and $s = l - h$ in eq. (4.2), we get

$$C\tau_s(l + h) = \Lambda_\Gamma(l + h) + C\kappa(l + h) \int_{l+h}^L \kappa(s')\tau(s')ds', \quad \text{and} \quad (4.4)$$

$$C\tau_s(l - h) = \Lambda_\Gamma(l - h) + C\kappa(l - h) \int_{l-h}^L \kappa(s')\tau(s')ds'. \quad (4.5)$$

Now,

$$\int_{l-h}^L \kappa(s')\tau(s')ds' = \int_{l+h}^L \kappa(s')\tau(s')ds' = J(l).$$

Substituting the expression of $J(l)$ from eq. (4.5) into eq. (4.4) and simplifying, we get

$$\tau_{2,s}(l)\kappa_1(l) - \tau_{1,s}(l)\kappa_2(l) = \frac{1}{C} \left[\Lambda_{\Gamma,2}(l)\kappa_1(l) - \Lambda_{\Gamma,1}(l)\kappa_2(l) \right].$$

Thus, the four boundary conditions are:

$$\tau(0) = \tau_0, \quad (4.6)$$

$$\tau_s(L) = \frac{\Lambda_\Gamma(l)}{C}, \quad (4.7)$$

$$\llbracket \tau(l) \rrbracket = -\frac{B\llbracket \kappa_s(l) \rrbracket}{C}, \quad \text{and} \quad (4.8)$$

$$\tau_{2,s}(l)\kappa_1(l) - \tau_{1,s}(l)\kappa_2(l) = \frac{1}{C} \left[\Lambda_{\Gamma,2}(l)\kappa_1(l) - \Lambda_{\Gamma,1}(l)\kappa_2(l) \right]. \quad (4.9)$$

If either $\kappa_1(l)$ or $\kappa_2(l)$ is zero, then eq. (4.4) and eq. (4.5) can be used to get the boundary condition instead of eq. (4.9).

It is important to note that the discontinuity in curvature introduces a $\delta_s(s-l)$ contribution in $\kappa_{ss}(s)$, which corresponds to a concentrated moment acting at the interface.

In the present formulation, the magnetic torque $\tau(s)$ is assumed to exhibit at most a finite jump discontinuity at $s = l$. Consequently, its derivative $\tau_s(s)$ may contain terms proportional to $\delta(s-l)$, but cannot generate $\delta_s(s-l)$ contributions.

Therefore, while the magnetic actuation can compensate concentrated force contributions arising from the discontinuity, it cannot eliminate the concentrated moment associated with the jump in curvature. As a result, a finite concentrated moment remains at the interface whenever $\llbracket \kappa(l) \rrbracket \neq 0$.

4.3.3 General Solution in Piecewise Domains

Consider two channels of curvatures $\kappa_1(s)$ and $\kappa_2(s)$ of lengths l and $L-l$ respectively, such that the total length is L and $\theta_1(l) = \theta_2(l)$.

Then, for $s < l$, we have

$$f_1(s) = -\frac{EI}{2}\kappa_2^2(L)\kappa_1(s) - \frac{EI}{2}\kappa_1^3(s) - EI\kappa_{ss}(s) + C\kappa_1(s) \int_s^L \kappa(s')\tau(s') ds' - C\tau_s(s). \quad (4.10)$$

Setting $f_1(s) = 0$ and differentiating eq. (4.10), we get

$$\kappa_1\tau_{ss} - \kappa_{1,s}\tau_s + \kappa_1^3\tau = -\frac{EI}{C} \left(\kappa_1\kappa_{1,sss} - \kappa_{1,s}\kappa_{1,ss} + \kappa_1^3\kappa_{1,s} \right). \quad (4.11)$$

Thus, the general solution for eq. (4.10) is

$$\tau_1(s) = A_1 \sin [\theta_1(s) - \theta_1(0)] + B_1 \cos [\theta_1(s) - \theta_1(0)] - \frac{EI}{C} \int_0^s g_1(s') \sin [\theta_1(s) - \theta_1(s')] ds'. \quad (4.12)$$

Similarly, for $l < s$, we have

$$\tau_2(s) = A_2 \sin [\theta_2(s) - \theta_2(l)] + B_2 \cos [\theta_2(s) - \theta_2(l)] - \frac{EI}{C} \int_l^s g_2(s') \sin [\theta_2(s) - \theta_2(s')] ds'. \quad (4.13)$$

In the above expressions

$$g_i(s) = \frac{d}{ds} \left(\frac{\kappa_i^2}{2} + \frac{\kappa_{i,ss}}{\kappa_i} \right).$$

Using the boundary conditions derived earlier, we can find the expressions for A_1, B_1, A_2 , and B_2 .

Special Case: Presence of a Straight Segment

The general solutions from eq. (4.12) and eq. (4.13) assume that the corresponding curvature functions are nonzero within the segment of interest.

If either segment is straight, i.e.,

$$\kappa_i(s) = 0,$$

the governing differential equation (4.3) becomes singular and must be replaced by the original lateral force balance equation.

As derived in the continuous curvature case, the magnetic torque remains constant within the straight segment:

$$\tau_i(s) = \tau_0.$$

Accordingly, in piecewise channels containing straight portions, the magnetic torque is obtained by combining constant solutions in the straight segments with the general solutions from eq. (4.12) or eq. (4.13) in the curved segments, together with the boundary and interface conditions derived previously.

4.3.4 Solution for Circle–Circle Channel of Different Curvatures

We consider a channel formed by two circular segments of different curvatures K_1 and K_2 such that

$$\theta_1(s) = K_1 s + \theta(0) \quad \text{and} \quad \kappa_2(s) = K_2(s - l) + \theta_1(l).$$

The expression for the magnetic torque is

$$\tau(s) = \begin{cases} A_1 \sin K_1 s + B_1 \cos K_1 s & s < l \\ A_2 \sin K_2 s + B_2 \cos K_2 s & l < s, \end{cases}$$

The integration constants are obtained as

$$A_1 = \frac{\sec \Phi}{2C} \left[B(K_2^2 - K_1^2) \cos(K_2(L - l)) - 2BK_2^2 + 2C\tau_0 \sin \Phi \right],$$

$$B_1 = \tau_0,$$

$$A_2 = \frac{\sec \Phi}{2C} \left[\sin(K_2 L) \left(B(K_2^2 - K_1^2) \sin(K_1 l) + 2C\tau_0 \right) - 2BK_2^2 \cos(l(K_1 - K_2)) \right] \quad \text{and}$$

$$B_2 = \frac{\sec \Phi}{2C} \left[\cos(K_2 L) \left(B(K_2^2 - K_1^2) \sin(K_1 l) + 2C\tau_0 \right) - 2BK_2^2 \sin(l(K_1 - K_2)) \right],$$

where

$$\Phi = K_1 l + K_2(L - l).$$

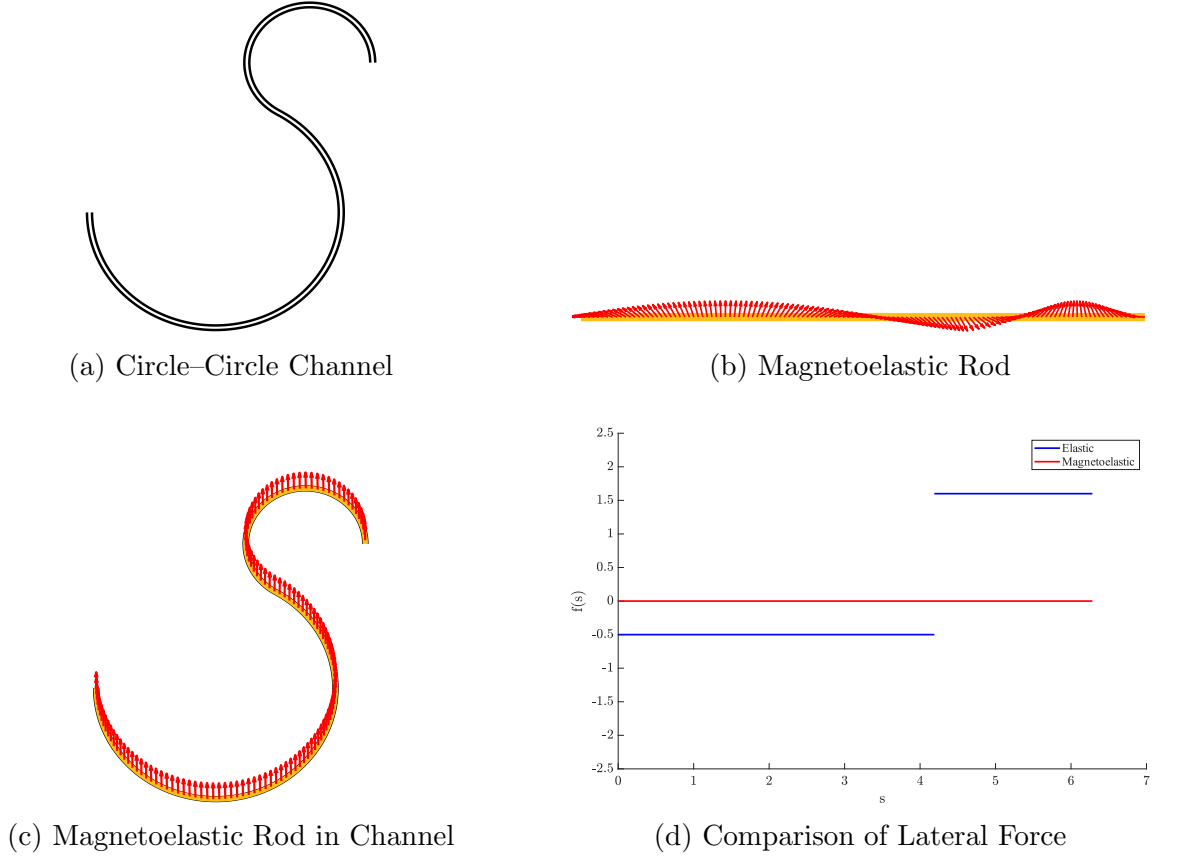


Figure 4.2: Comparison of elastic and magnetoelastic behavior in a circle-circle channel. (a) Prescribed channel geometry of radii $R_1 = 1$ and $R_2 = 0.5$ with $\theta(0) = -90^\circ$. (b) Magnetoelastic rod of length $L = \frac{4\pi R_1}{3} + \frac{4\pi R_2}{3}$ with the corresponding embedded magnetization distribution having $\gamma(0) = 0^\circ$ obtained for $H_{ext} = 0.45$ and $\beta = 90^\circ$. (c) Equilibrium configuration of the magnetoelastic rod within the channel. (d) Comparison of the lateral force, showing elimination of the lateral force in the magnetoelastic case.

4.4 Rod Insertion

4.4.1 Relation between Channel Curvatures

To solve the rod insertion problem, we need $\gamma(s) = \gamma_0$. From eq. (3.29), we know the channels that are compatible with rod insertion if

$$\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} = \text{constant}.$$

Let $\tau_1(s) = \sin(\beta - \theta_1(s) - \gamma_0)$ and $\tau_2(s) = \sin(\beta - \theta_2(s) - \gamma_0)$. Now,

$$\begin{aligned}\tau_1(s) &= \sin(\beta - \theta_1(s) - \gamma_0) \\ &= \sin(\beta - \theta_1(0) - \gamma_0) \cos[\theta_1(s) - \theta_1(0)] - \cos(\beta - \theta_1(0) - \gamma_0) \sin[\theta_1(s) - \theta_1(0)]\end{aligned}$$

Comparing with eq. (4.12), we have

$$\begin{aligned}B_1 &= \tau_0, \quad \text{and} \\ A_1 &= -\cos(\beta - \theta_1(0) - \gamma_0)\end{aligned}$$

Now,

$$\begin{aligned}\tau_2(s) &= \sin(\beta - \theta_2(s) - \gamma_0) \\ &= \sin(\beta - \theta_2(l) - \gamma_0) \cos[\theta_2(s) - \theta_2(l)] - \cos(\beta - \theta_2(l) - \gamma_0) \sin[\theta_2(s) - \theta_2(l)].\end{aligned}$$

Compared with eq. (4.13), we have

$$\begin{aligned}B_2 &= \sin(\beta - \theta_2(l) - \gamma_0) \\ &= B_1 \cos[\theta_2(l) - \theta_1(0)] + A_1 \sin[\theta_2(l) - \theta_1(0)], \quad \text{and} \\ A_2 &= -\cos(\beta - \theta_2(l) - \gamma_0) \\ &= A_1 \cos[\theta_2(l) - \theta_1(0)] - B_1 \sin[\theta_2(l) - \theta_1(0)].\end{aligned}$$

To find the expression for A_1 , we will use the boundary conditions. It can be shown that eq. (4.6) and eq. (4.8) are automatically satisfied. Using eq. (4.7), we get

$$EI \left(\frac{\kappa_1^2(l)}{2} + \frac{\kappa_{1,ss}(l)}{\kappa_1(l)} \right) - EI \left(\frac{\kappa_2^2(l)}{2} + \frac{\kappa_{2,ss}(l)}{\kappa_2(l)} \right) = 0.$$

Thus,

$$\frac{\kappa_1^2(l)}{2} + \frac{\kappa_{1,ss}(l)}{\kappa_1(l)} = \frac{\kappa_2^2(l)}{2} + \frac{\kappa_{2,ss}(l)}{\kappa_2(l)}.$$

We know

$$\frac{\kappa_1^2(s)}{2} + \frac{\kappa_{1,ss}(s)}{\kappa_1(s)} = \alpha_1, \quad \frac{\kappa_2^2(s)}{2} + \frac{\kappa_{2,ss}(s)}{\kappa_2(s)} = \alpha_2.$$

Thus, to solve the rod insertion problem (when $k_1(s)$ and $k_2(s)$ are not zero)

$$\frac{\kappa_1^2(s)}{2} + \frac{\kappa_{1,ss}(s)}{\kappa_1(s)} = \frac{\kappa_2^2(s)}{2} + \frac{\kappa_{2,ss}(s)}{\kappa_2(s)} = \alpha. \quad (4.14)$$

4.4.2 Expression for Constant Magnetization Angle

From eq. (4.9), we have

$$A_2 \cos [\theta_2(L) - \theta_2(l)] - B_2 \sin [\theta_2(L) - \theta_2(l)] = -\frac{EI}{C} \left(\kappa_2^2(L) + \frac{\kappa_{2,ss}(L)}{\kappa_2(L)} \right).$$

Substituting the expression for A_2 and B_2 , we get

$$A_1 = \tau_0 \tan [\theta_2(L) - \theta_1(0)] - \frac{EI \left(\frac{\kappa_2^2(L)}{2} + \alpha \right)}{C \cos [\theta_2(L) - \theta_1(0)]}.$$

Also,

$$\cos(\beta - \theta_2(L) - \gamma_0) = \frac{EI}{C} \left(\frac{\kappa_2^2(L)}{2} + \alpha \right).$$

Thus,

$$\gamma_0 = \beta - \theta_2(L) \pm \cos^{-1} \left[\frac{EI}{C} \left(\frac{\kappa_2^2(L)}{2} + \alpha \right) \right]. \quad (4.15)$$

If $\kappa_1(s) = 0$, the expression for γ_0 remains the same with α evaluated using $\kappa_2(s)$.

Similarly, if $\kappa_2(s) = 0$, then α is evaluated using $\kappa_1(s)$ and the expression for γ_0 remains the same.

4.5 Rod Insertion Solutions for Canonical Curvatures

4.5.1 Straight and Circle

Considering $\kappa_1(s) = 0$ and $\kappa_2(s) = K$, we the expression for constant magnetization angle as

$$\gamma_0 = \beta - \theta(0) - K(L - l) \pm \cos^{-1} \left(\frac{EIK^2}{C} \right).$$

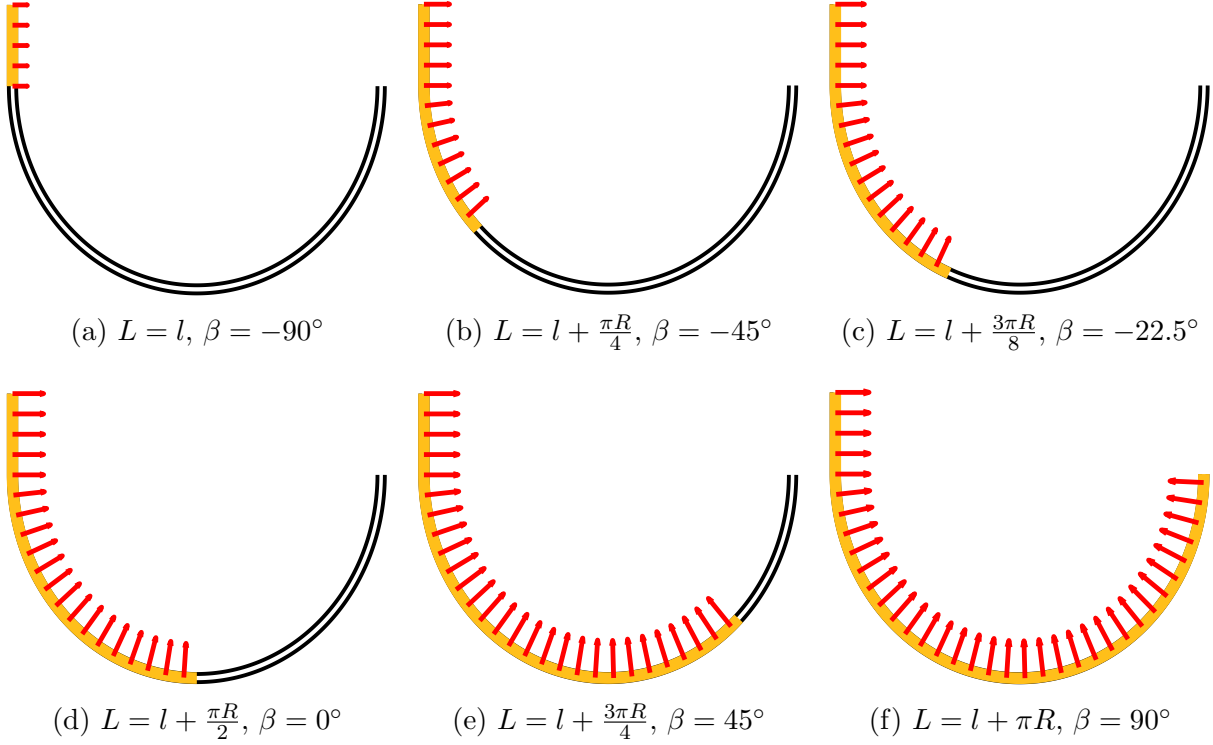


Figure 4.3: Quasi-static insertion of a uniformly magnetized rod into a straight–circular channel, where the length of the straight segment is $l = 0.2$ and radius of circular segment is $R = 0.5$. Figures (a)–(f) show successive equilibrium configurations for increasing inserted length, with the external magnetic field orientation β adjusted to maintain the constant embedded magnetization angle throughout insertion at $H_{ext} = 0.45$.

4.5.2 Circle and Circle

Using eq. (4.14), we get

$$\kappa_1^2(s) = \kappa_2^2(s).$$

Let $\kappa_1(s) = K_1$ and $\kappa_2(s) = K_2$. Thus, for rod insertion, we have

$$K_2 = \pm K_1.$$

Using eq. (4.15), we get the expression for constant magnetization angle as

$$\gamma_0 = \beta - \theta(0) - K_1 l - K_2 (L - l) \pm \cos^{-1} \left(\frac{E I K_2^2}{C} \right).$$

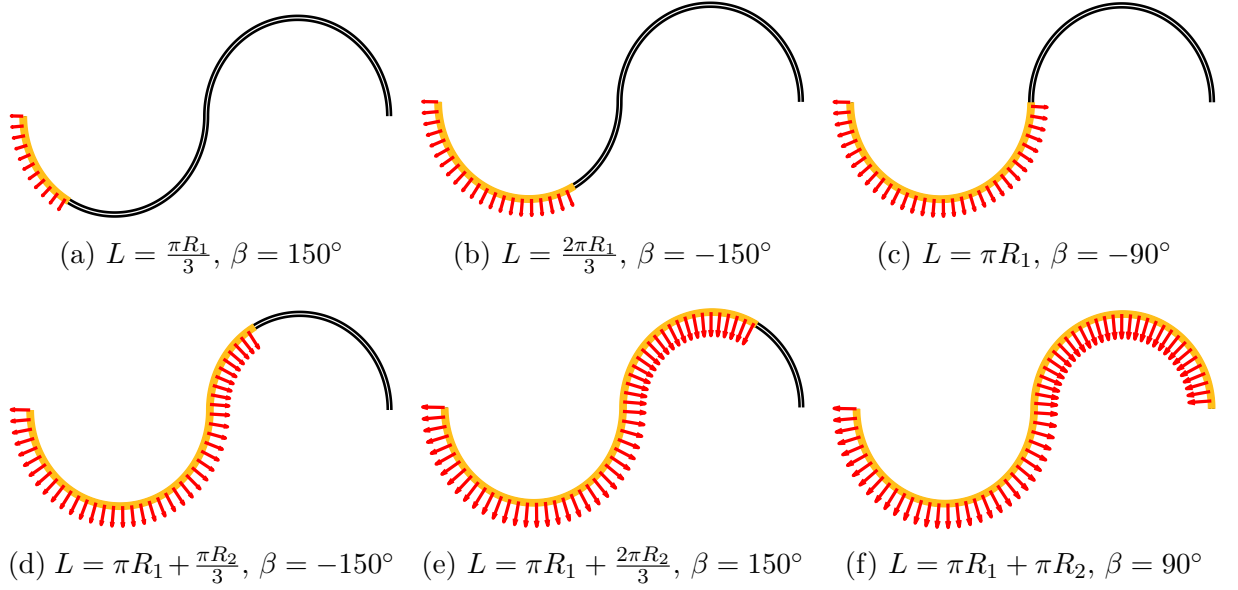


Figure 4.4: Quasi-static insertion of a uniformly magnetized rod into a circle-circle channel with radii $R_1 = R_2 = 0.5$ and having opposite curvature. Figures (a)–(f) show successive equilibrium configurations for increasing inserted length, with the external magnetic field orientation β adjusted to maintain the constant embedded magnetization angle throughout insertion at $H_{ext} = 0.45$.

4.5.3 Intestinal-Type Geometries

The discontinuous-curvature framework developed in the preceding sections can also be employed to construct geometries resembling confined biological pathways. In particular, planar approximations of intestinal-type channels may be generated using piecewise curvature segments connected continuously while allowing finite curvature discontinuities at the interfaces.

4.5.3.1 Piecewise Circular Approximation

A simple intestinal-type geometry may be constructed using straight and circular segments. Within each segment, the curvature is piecewise constant and is given by

$$\kappa(s) = K_i, \quad s \in [s_{i-1}, s_i],$$

where K_i denotes the curvature of the i^{th} segment and $l_i = s_i - s_{i-1}$ denotes the corresponding segment length.

Straight portions correspond to

$$K_i = 0,$$

while curved portions satisfy

$$K_i = \pm K,$$

depending on the direction of bending.

The tangent angle is obtained from

$$\theta(s) = \theta(s_{i-1}) + K_i(s - s_{i-1}), \quad s \in [s_{i-1}, s_i],$$

while continuity of $\theta(s)$ is enforced across neighboring segments.

Using the constant-magnetization-angle condition derived previously, the admissible value of γ_0 is obtained as

$$\gamma_0 = \beta - \sum_{i=1}^N K_i l_i \pm \cos^{-1} \left(\frac{EIK^2}{C} \right). \quad (4.16)$$

The resulting piecewise circular geometry provides a simple canonical example for studying the influence of curvature discontinuities and the associated interface effects.

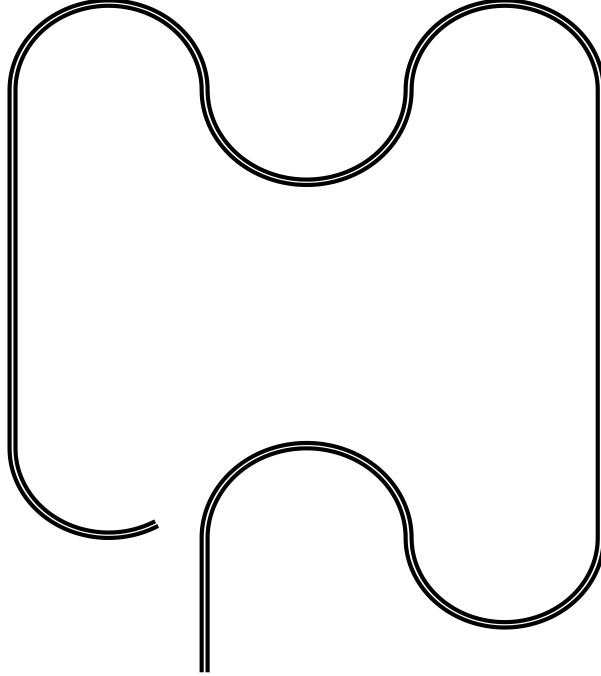


Figure 4.5: Piecewise straight-circular approximation of an intestinal-type channel geometry within the discontinuous curvature formulation.

4.5.3.2 Piecewise Elastica Approximation

To construct better intestinal-like geometries within the discontinuous-curvature framework that allow rod insertion, piecewise curvature segments based on Jacobi elliptic functions were considered. Within each segment, the curvature was chosen in the form given by eq. (3.32). The shape of the different segments is obtained by varying a and m , such that eq. (4.14) is satisfied for all the segments to ensure rod insertion.

The tangent angle within each segment is obtained from

$$\theta(s) = \theta(s_{i-1}) + \int_{s_{i-1}}^s k(s') ds',$$

while continuity of $\theta(s)$ is enforced across neighboring segments. The resulting geometry is shown in fig. 4.6. Different colors correspond to different curvature segments used in the construction. This approximation reduces the number of points of discontinuous curvature while giving a better approximation.

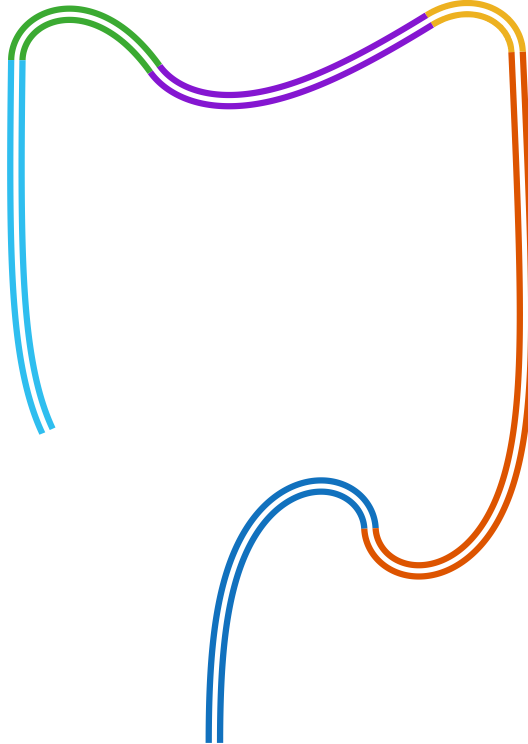


Figure 4.6: Intestinal-type geometry generated using piecewise elastica segments. Different colors indicate different elastica-based curvature segments used in the construction.

Chapter 5

Direct Problem

In the preceding chapters, the inverse problem was considered, in which the channel geometry was prescribed, and the corresponding magnetic actuation required to eliminate lateral forces was determined. This formulation establishes the relationship between a desired trajectory and the actuation necessary for the rod to conform to it.

We now consider the complementary *direct problem*, in which the external loads and magnetic torque are prescribed, and the resulting configuration of the rod is determined. The objective is also to verify whether the loading conditions obtained from the inverse problem reproduce the same channel geometry when applied to the rod. This analysis also serves as a consistency check for the formulation and establishes that the inverse and direct problems are mutually compatible.

5.1 Continuous Curvature

5.1.1 Semi-Circular Channel

We consider a channel of constant curvature $\kappa(s) = K$. The objective is to verify that the loading conditions obtained from the inverse problem reproduce the same configuration when applied in the direct formulation.

5.1.1.1 Elastic Rod

For the elastic case, the boundary loads obtained from the inverse formulation are

$$\begin{aligned} N(L) &= -EIK^2, & H(L) &= 0, \\ N(0) &= -EIK^2, & H(0) &= 0. \end{aligned}$$

Thus,

$$\mathbf{n}(L) = -EIK^2 \mathbf{d}_3(L) = -EIK^2 \mathbf{e}_1, \quad \mathbf{n}(0) = -EIK^2 \mathbf{d}_3(0) = EIK^2 \mathbf{e}_1.$$

The distributed force acting on the rod is

$$\mathbf{f}(s) = EIK^3 \hat{\mathbf{f}}(s),$$

where $\hat{\mathbf{f}}(s) = \cos[\theta(s) - \theta(0)] \mathbf{e}_3 + \sin[\theta(s) - \theta(0)] \mathbf{e}_1$.

The equilibrium equations are given by

$$\begin{aligned} \mathbf{n}_s + \mathbf{f}(s) &= \mathbf{0}, \\ EI\theta_{ss} + (\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 &= 0. \end{aligned}$$

Integrating the force balance equation from s to L ,

$$\mathbf{n}(s) = \mathbf{n}(L) + \int_s^L \mathbf{f}(s') ds'.$$

Substituting the expression, we obtain

$$\mathbf{n}(s) = -EIK^2 \sin(Ks) \mathbf{e}_3 + EIK^2 \cos(Ks) \mathbf{e}_1.$$

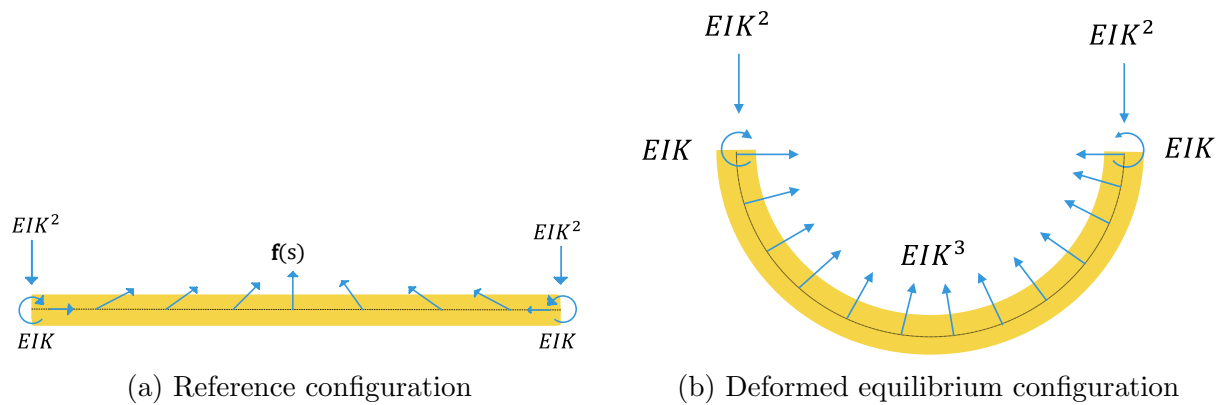


Figure 5.1: Direct problem formulation: (a) loading applied to the elastic rod in its reference configuration, including the distributed load $f(s)$ and the boundary forces and moments, and (b) the corresponding deformed equilibrium configuration obtained under the applied loading.

Substituting $\mathbf{r}_s = \cos \theta \mathbf{e}_3 + \sin \theta \mathbf{e}_1$ we get

$$(\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 = EIK^2 \cos(\theta - Ks).$$

Thus, the governing equation becomes

$$EI\theta_{ss}(s) + EIK^2 \cos(\theta - Ks) = 0,$$

with $\theta_s(0) = \theta_s(L) = K$.

Requiring this equation to hold for all K leads to

$$\theta(s) - Ks = 0.$$

This implies

$$\theta(s) = Ks + C.$$

Using the boundary condition $\theta(0) = \frac{3\pi}{2}$, we obtain

$$\theta(s) = Ks + \frac{3\pi}{2}.$$

This matches the geometry of the prescribed channel, confirming that the direct problem recovers the desired configuration.

5.1.1.2 Magnetoelastic Rod

For the magnetoelastic case, the equilibrium equations are

$$\begin{aligned} \mathbf{n}_s + \mathbf{f}(s) &= \mathbf{0}, \\ EI\theta_{ss} + (\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 + C\tau &= 0. \end{aligned}$$

In the direct problem, the magnetic torque distribution is not prescribed independently. Instead, the magnetization angle $\gamma(s)$ obtained from the inverse formulation is assumed to be embedded within the initially straight rod in the reference frame, and the magnetic torque $\tau_{ref}(s)$ acts initially with

$$\tau_{ref}(s) = \sin(\beta - \gamma(s)).$$

Upon application of the external magnetic field and boundary loads, the rod deforms toward an equilibrium configuration characterized by the tangent angle $\theta(s)$. At

equilibrium, the resulting magnetic torque distribution is given by

$$\tau_{eq}(s) = \sin(\beta - \theta(s) - \gamma(s)).$$

The expressions for $\tau_{eq}(s)$ derived previously in the inverse problem therefore represent the equilibrium magnetic torque distributions corresponding to the prescribed magnetization profile. In the present direct formulation, these torque distributions are substituted into the equilibrium equations to verify that the resulting deformed configuration reproduces the prescribed channel geometry.

For the prescribed magnetization distribution corresponding to a circular channel, the resulting equilibrium magnetic torque distribution is

$$\tau_{eq}(s) = \frac{EIK^2}{C} \sin(Ks) + \tau_0 \cos(Ks).$$

For the magnetoelastic case, the boundary loads obtained from the inverse formulation are

$$\begin{aligned} N(L) &= -EIK^2, & H(L) &= C\tau_0, \\ N(0) &= EIK^2, & H(0) &= -C\tau_0. \end{aligned}$$

Thus,

$$\begin{aligned} \mathbf{n}(L) &= -EIK^2 \mathbf{d}_3(L) + C\tau(0) \mathbf{d}_1(L) \\ &= -EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3. \end{aligned}$$

In this case, $\mathbf{f}(s) = \mathbf{0}$, so the internal force is constant:

$$\mathbf{n} = \mathbf{n}(L) = \mathbf{n}(0) = -EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3.$$

Substituting into the moment balance equation,

$$(\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 = -EIK^2 \cos \theta + C\tau_0 \sin \theta.$$

Thus, the governing equation becomes

$$EI\theta_{ss} - EIK^2 \cos \theta(s) + C\tau_0 \sin \theta(s) + EIK^2 \sin(Ks) + C\tau_0 \cos(Ks) = 0,$$

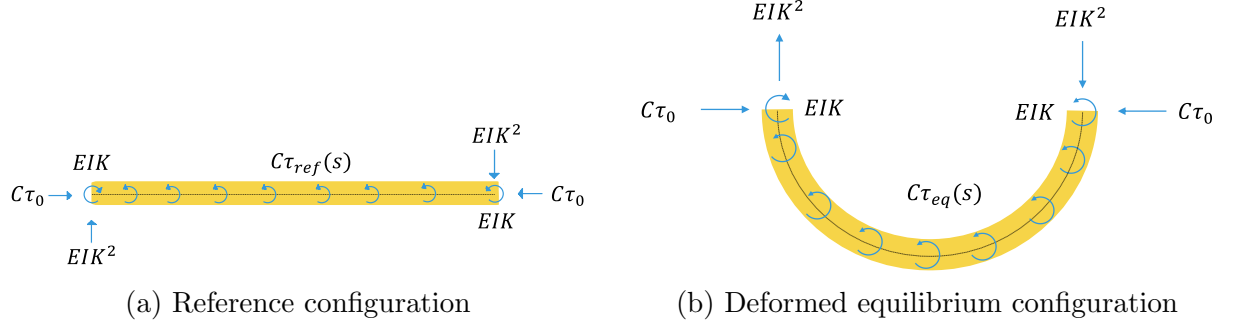


Figure 5.2: Direct problem formulation for the magnetoelastic rod: (a) reference configuration with the distributed magnetic torque $\tau_{ref}(s)$ and the boundary forces and moments required to produce the prescribed deformation, and (b) the corresponding deformed equilibrium configuration with the transformed magnetic torque distribution $\tau_{eq}(s)$.

with $\theta_s(0) = \theta_s(L) = K$.

Requiring this equation to hold for all admissible values of $\tau(0)$ leads to

$$\sin \theta + \cos(Ks) = 0.$$

This implies

$$\theta(s) = Ks + \frac{3\pi}{2},$$

which satisfies both the governing equation and the boundary conditions.

Thus, the magnetoelastic rod also reproduces the prescribed circular geometry under the applied loading.

5.2 Discontinuous Curvature

The direct problem for a purely elastic rod was analyzed for continuously varying curvature. Since the underlying procedure remains unchanged in the present case, the elastic formulation is not repeated here.

Instead, attention is focused on the magnetoelastic case, where the presence of curvature discontinuities and interface conditions leads to additional mechanical effects not encountered in the continuous-curvature setting.

In the discontinuous-curvature setting, the direct problem is solved separately in each channel segment, with jump conditions imposed at the interface as loads.

The objective is to verify that the loading conditions obtained from the inverse formulation reproduce the prescribed piecewise geometry while satisfying the interface

conditions associated with the curvature discontinuity.

5.2.1 Straight and Semi-circle

For the magnetoelastic case, the equilibrium equations are

$$\begin{aligned}\mathbf{n}_s + \mathbf{f} &= \mathbf{0} \\ EI\theta_{ss} + (\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 + C\tau &= 0\end{aligned}$$

For a channel consisting of a straight segment followed by a semicircular segment, the equilibrium magnetic torque for the prescribed magnetization angle is given as

$$\tau_{eq}(s) = \begin{cases} \tau_0 & s < l \\ A \sin Ks + B \cos Ks & l < s, \end{cases}$$

where

$$\begin{aligned}A &= \tau_0 \sin Kl - \frac{\tau_s(L)}{K} \cos Kl, \quad \text{and} \\ B &= \tau_0 \cos Kl + \frac{\tau_s(L)}{K} \sin Kl\end{aligned}$$

The boundary conditions obtained from the inverse problem are

$$\begin{aligned}N(L) &= -EIK^2, \quad H(L) = C\tau_0, \\ N(0) &= \frac{3}{2}EIK^2, \quad H(0) = -C\tau_0.\end{aligned}$$

Thus,

$$\begin{aligned}\mathbf{n}(L) &= -EIK^2 \mathbf{d}_3(L) + C\tau_0 \mathbf{d}_1(L) \\ &= -EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3.\end{aligned}$$

Similarly,

$$\begin{aligned}\mathbf{n}(0) &= -\frac{3}{2}EIK^2 \mathbf{d}_3(0) - C\tau_0 \mathbf{d}_1(0) \\ &= \frac{3}{2}EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3.\end{aligned}$$

Although $\mathbf{f}(s)$ is zero in the distribution, due to the presence of a discontinuity in the curvature, a point load and a concentrated moment act at the point of discontinuity, as discussed earlier. Thus,

$$\begin{aligned}\mathbf{f}(s) &= \frac{EI}{2}K^2\delta(s-l)\mathbf{d}_3(l) - EIK\delta_s(s-l)\mathbf{d}_2(l) \\ &= -\frac{EI}{2}K^2\delta(s-l)\mathbf{e}_1 - EIK\delta_s(s-l)\mathbf{e}_2.\end{aligned}$$

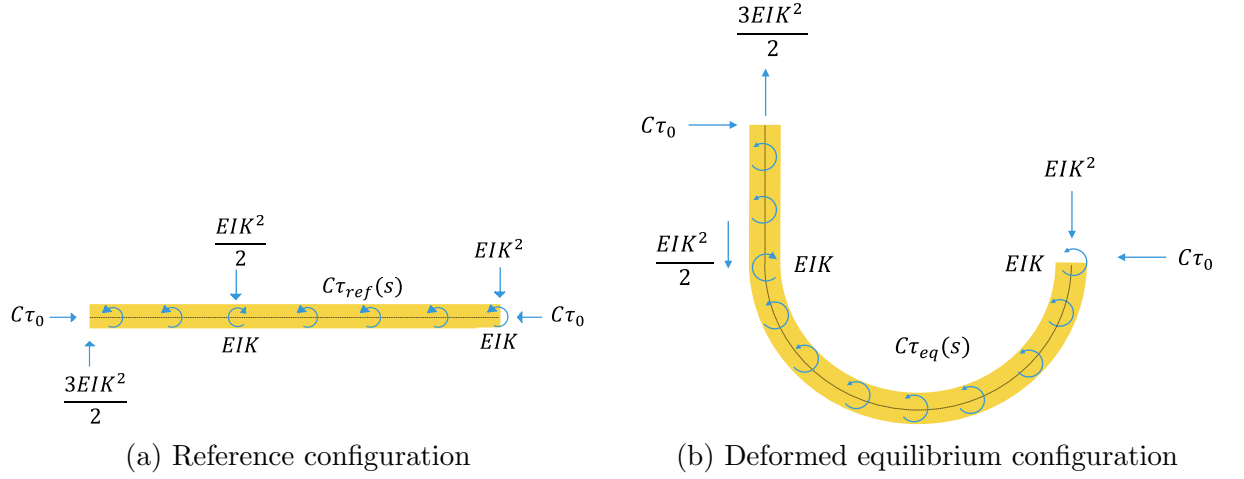


Figure 5.3: Direct problem formulation for the straight-semi-circular channel example: (a) reference configuration with the distributed magnetic torque $\tau_{\text{ref}}(s)$, the required boundary forces and moments, and the concentrated internal loading arising from the curvature discontinuity, and (b) the corresponding deformed equilibrium configuration with the transformed magnetic torque distribution $\tau_{\text{eq}}(s)$.

Straight Segment: $s < l$

The integration of the force balance equation yields

$$\mathbf{n}(s) = -\frac{3}{2}EIK^2\mathbf{e}_1 - C\tau_0\mathbf{e}_3,$$

and

$$(\mathbf{r}_s(s) \times \mathbf{n}(s)) \cdot \mathbf{e}_2 = -\frac{3}{2}EIK^2 \cos \theta + C\tau_0 \sin \theta.$$

Thus, the governing differential equation becomes

$$EI\theta_{ss} - \frac{3}{2}EIK^2 \cos \theta + C\tau_0 \sin \theta + C\tau_0 = 0,$$

with $\theta_s(0) = \theta_s(l^-) = 0$.

For $\theta(s)$ to be a solution for all admissible τ_0 , we have

$$\sin \theta + 1 = 0 \implies \theta(s) = (2n + 1) \frac{\pi}{2}$$

Since $\theta(0) = 3\frac{\pi}{2}$, we have $n = 1$. Thus,

$$\theta(s) = 3\frac{\pi}{2}.$$

Semi-circular Segment: $l < s$

The integration of the force balance equation yields

$$\mathbf{n}(s) = -EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3.$$

Thus,

$$(\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 = -EIK^2 \cos \theta + C\tau_0 \sin \theta.$$

The governing differential equation becomes

$$EI\theta_{ss} - EIK^2 \cos \theta + C\tau_0 \sin \theta + EIK^2 \sin Ks + C\tau_0 \cos Ks = 0,$$

with $\theta_s(l^+) = \theta_s(L) = K$.

For $\theta(s)$ to be a solution for all admissible τ_0 , we have

$$\sin \theta + \cos(Ks - Kl) = 0 \implies \theta(s) = K(s - l) + \frac{\pi}{2} + 2n\pi$$

Since $\theta(l) = 3\frac{\pi}{2}$, we have $n = 1$. Thus,

$$\theta(s) = K(s - l) + 3\frac{\pi}{2}.$$

5.2.2 Semi-Circle and Semi-Circle

For the magnetoelastic case, the equilibrium equations are

$$\mathbf{n}_s + \mathbf{f} = \mathbf{0}$$

$$EI\theta_{ss} + (\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 + C\tau = 0$$

For a channel consisting of two semi-circular segments of opposite curvatures, the equilibrium magnetic torque corresponding to the prescribed magnetization angle is given as

$$\tau_{eq}(s) = \begin{cases} -\frac{EIK^2}{C} \sin Ks + \tau_0 \cos Ks, & s < l \\ \frac{EIK^2}{C} \sin Ks + \tau_0 \cos Ks, & l < s. \end{cases}$$

The boundary loads obtained from the inverse problem are

$$\begin{aligned} N(L) &= -EIK^2, H(L) = -C\tau_0, \\ N(0) &= -EIK^2, H(0) = -C\tau_0. \end{aligned}$$

Thus,

$$\begin{aligned} \mathbf{n}(L) &= -EIK^2 \mathbf{d}_3(L) - C\tau_0 \mathbf{d}_1(L) \\ &= EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3. \end{aligned}$$

Similarly,

$$\begin{aligned} \mathbf{n}(0) &= -EIK^2 \mathbf{d}_3(0) - C\tau_0 \mathbf{d}_1(0) \\ &= EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3. \end{aligned}$$

In this case, $\mathbf{f}(s)$ is zero in the distribution and also does not have a point load component since $\llbracket k_s(l) \rrbracket = 0$. But, there still is a concentrated moment acting at the point of discontinuity. Hence,

$$\mathbf{f}(s) = -2EIK\delta_s(s-l)\mathbf{d}_2(l) = -2EIK\delta_s(s-l)\mathbf{e}_2.$$

Semi-circular Segment 1: $s < l$

The integration of the force balance equation yields

$$\mathbf{n}(s) = EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_3,$$

and

$$(\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 = EIK^2 \cos \theta + C\tau_0 \sin \theta.$$

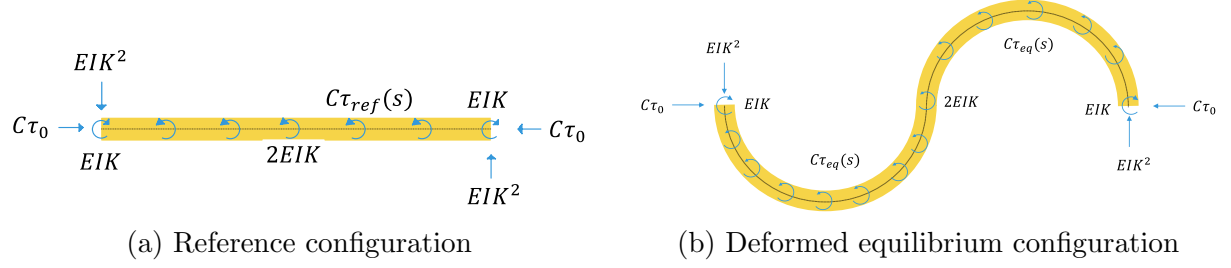


Figure 5.4: Direct problem formulation for the semi-circular–semi-circular discontinuous-curvature example: (a) reference configuration with the distributed magnetic torque $\tau_{\text{ref}}(s)$, the required boundary forces and moments, and the concentrated internal loading at the curvature discontinuity, and (b) the corresponding deformed equilibrium configuration with the transformed magnetic torque distribution $\tau_{\text{eq}}(s)$.

Thus, the governing equation simplifies to

$$EI\theta_{ss} + EIK^2 \cos \theta + C\tau_0 \sin \theta - EIK^2 \sin Ks + C\tau_0 \cos Ks = 0,$$

with $\theta_s(0) = \theta_s(l^-) = K$. For $\theta(s)$ to be a solution for all admissible τ_0 , we have

$$\sin \theta + \cos(Ks) = 0 \implies \theta(s) = Ks - \frac{\pi}{2} + 2n\pi.$$

Since $\theta(0) = 3\frac{\pi}{2}$, we have $n = 1$. Thus,

$$\theta(s) = Ks + 3\frac{\pi}{2}.$$

Semi-circular Segment 2: $l < s$

Integrating the force balance equation yields

$$\mathbf{n}(s) = EIK^2 \mathbf{e}_1 - C\tau_0 \mathbf{e}_1,$$

and

$$(\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_2 = EIK^2 \cos \theta + C\tau_0 \sin \theta.$$

The governing differential equation simplifies to

$$EI\theta_{ss} + EIK^2 \cos \theta + C\tau_0 \sin \theta + EIK^2 \sin Ks + C\tau_0 \cos Ks = 0,$$

with $\theta_s(l^+) = \theta_s(L) = -K$. For $\theta(s)$ to be a solution for all admissible τ_0 , we have

$$\sin \theta + \cos(Ks) = 0 \implies \theta(s) = -Ks - \frac{\pi}{2} + 2n\pi.$$

Since $\theta(l) = Kl_1 + 3\frac{\pi}{2} = 5\frac{\pi}{2}$, we have $n = 2$. Thus,

$$\begin{aligned} \theta(s) &= -Ks - \frac{\pi}{2} + 4\pi \\ &= -Ks - \frac{\pi}{2} + Kl + Kl + 2\pi \\ &= -K(s - l) + Kl + 3\frac{\pi}{2}. \end{aligned}$$

These results show that for a given channel geometry, a corresponding magnetic actuation can be determined, and conversely, that the loading conditions with magnetic actuation reproduce the same geometry. Thus, the formulation provides a consistent framework for both determining the required actuation for a prescribed path and predicting the resulting rod configuration.

5.3 Physical Interpretation

The solution of the direct problem demonstrates that the loading conditions derived from the inverse formulation are sufficient to reproduce the prescribed channel geometry. More generally, the direct formulation shows that by appropriately selecting the boundary forces, boundary moments, and distributed magnetic actuation, a wide class of desired equilibrium configurations can be generated for the magnetoelastic rod.

Chapter 6

Conclusions and Future Work

6.1 Summary of Results

6.1.1 Continuous Curvature

In this work, the mechanics of a slender rod constrained within channels of prescribed continuous curvature was studied using the planar Cosserat rod framework. For the purely elastic case, it was shown that constraining the rod to follow an arbitrary channel geometry generally produces nonzero lateral forces due to curvature-induced internal stresses.

The formulation was subsequently extended to magnetoelastic rods by incorporating a distributed magnetic torque within the moment balance equation. A general analytical framework was developed to determine the magnetic torque distribution required to eliminate the lateral force acting on the rod.

A governing differential equation for the magnetic torque distribution was obtained, together with the corresponding boundary conditions. Analytical solutions were derived for arbitrary curvature distributions and subsequently specialized to representative geometries including circular, sinusoidal, and spiral channels. These examples demonstrated the generality of the formulation and illustrated the influence of curvature variation on the required magnetic actuation.

The relationship between magnetic torque and the embedded magnetization distribution was established, enabling reconstruction of admissible magnetization profiles for prescribed channel geometries and identification of special constant-magnetization-angle solutions relevant to rod insertion.

6.1.2 Discontinuous Curvature

The analysis was extended to channels with discontinuous curvature, where the curvature exhibits finite jump discontinuities at isolated points. While the governing equilibrium equations remain valid within each smooth segment, additional interface conditions arise due to the discontinuity in curvature.

For the purely elastic rod, jump conditions in the axial force, shear force, and bending moment were derived. The curvature discontinuity was shown to introduce singular contributions within the curvature derivatives, leading to concentrated mechanical effects at the interface.

The magnetoelastic formulation was subsequently developed for piecewise-curvature channels. It was shown that the lateral force can again be eliminated through appropriate selection of the magnetic torque distribution and magnetic torque. Analytical solutions were obtained for representative discontinuous-curvature geometries, including straight–circle and circle–circle transitions.

A key result of the discontinuous-curvature analysis is the identification of singular interface effects arising from the curvature jump. In particular, the curvature derivatives contain terms proportional to Dirac delta and derivative-of-delta distributions. While the magnetic actuation can compensate concentrated force contributions associated with $\delta(s - l)$ terms, it cannot eliminate the concentrated moments associated with $\delta'(s - l)$ contributions when the magnetic torque field is restricted to at most finite jump discontinuities.

The formulation was further extended to complex piecewise-curvature geometries consisting of alternating straight and curved segments. It was shown that admissible constant- γ solutions can exist even for geometries containing multiple curvature discontinuities.

6.1.3 Direct and Inverse Problems

Two complementary formulations were investigated in this work.

In the inverse problem, the channel geometry is prescribed, and the objective is to determine the magnetic torque distribution required for the rod to conform to the channel while maintaining equilibrium.

In the direct problem, the loading and magnetic actuation are prescribed, and the resulting rod configuration is determined. It was demonstrated that the loading conditions and magnetic torque distributions obtained from the inverse formulation reproduce the

original prescribed channel geometry when applied within the direct formulation.

This establishes a direct correspondence between geometry and actuation, showing that appropriate magnetic loading can be designed to guide the rod along desired paths. The equivalence between the direct and inverse formulations also provides a consistent framework for both analysis and design of magnetically actuated rod insertion systems.

6.2 Key Insights

6.2.1 Elimination of Lateral Force

It was shown that the lateral force can again be eliminated through the appropriate selection of the magnetic torque distribution. By appropriately selecting the magnetic torque distribution, the internal elastic forces generated by the channel curvature can be counterbalanced through magnetic actuation.

This provides a systematic mechanism for guiding slender rods through confined pathways while minimizing contact forces with the surrounding boundaries.

6.2.2 Role of Curvature Discontinuities

A clear distinction exists between continuously varying and discontinuous channel geometries. A governing differential equation characterizing the magnetic torque distribution required for elimination of the lateral force was obtained, together with the corresponding boundary conditions. The governing equations remain smooth, and the actuation problem does not involve singular interface effects.

In contrast, curvature discontinuities introduce singular interface effects and discontinuities in the internal force components. The resulting jump conditions impose additional compatibility requirements on the admissible solutions and significantly influence the required actuation strategy.

The distributional interpretation of the curvature derivatives further clarifies the physical origin of concentrated interface forces and moments.

6.2.3 Constant Magnetization Angle Solutions

A special class of solutions corresponding to a constant angle between the embedded magnetization vector and the chosen reference direction was identified and analyzed. These constant- γ solutions provide simplified actuation strategies and establish direct relationships between geometry and magnetic loading.

Conditions governing the existence of such solutions were derived for both continuous

and discontinuous curvature profiles. The resulting framework enables the construction of admissible piecewise-curvature geometries supporting constant magnetization angle actuation.

The constant-magnetization-angle framework further provides a quasi-static insertion strategy in which the embedded magnetization remains fixed while the external field orientation and holding force are adjusted continuously as insertion progresses.

Special attention was given to degenerate configurations in which either the terminal curvature vanished or the boundary conditions lost sensitivity to the homogeneous solution component. In such cases, the governing boundary value problem no longer uniquely determines the magnetic torque distribution.

It was shown, however, that the constraint

$$|\tau(s)| \leq 1,$$

imposes admissibility restrictions on the solution family, thereby constraining the allowable homogeneous contributions.

This distinction between uniqueness imposed by boundary conditions and admissibility imposed by constitutive physics provides additional insight into the mathematical structure of the formulation.

6.3 Limitations

The present formulation is developed under a number of simplifying assumptions that define its scope and applicability.

The analysis is restricted to quasi-static planar equilibrium, and therefore, does not account for inertial effects or time-dependent actuation. As a result, the insertion strategies developed here describe equilibrium transitions between configurations rather than the full dynamics of insertion.

The rod is modeled as an inextensible and unshearable Cosserat rod with linear elastic bending response. Material nonlinearities, torsional deformation, axial extensibility, and shear deformation are neglected. While these assumptions are appropriate for slender rods undergoing moderate planar bending, they may not remain valid for highly compliant structures or strongly nonlinear material behavior.

The channel is assumed perfectly rigid and frictionless, with the rod fully constrained to the prescribed centerline. In practical insertion problems, wall deformation and frictional contact with the channel may significantly alter the force transmission and

equilibrium behavior.

Although the formulation enables elimination of the distributed lateral force acting along the rod through appropriate magnetic actuation, this elimination is achieved only in the distributional sense. Boundary reaction forces and moments remain necessary to maintain equilibrium, and therefore, the rod is not entirely free of mechanical loading. In discontinuous-curvature channels, concentrated interface effects associated with curvature jumps may also persist, particularly concentrated moments arising from derivative-of-delta contributions that cannot be compensated within the present actuation framework. Thus, the present formulation eliminates distributed lateral contact loading but does not eliminate all contact or reaction forces within the system.

The magnetic actuation model is restricted to planar magneto-mechanical interactions, consistent with the planar Cosserat rod formulation adopted in this work. Consequently, out-of-plane deformations, torsional effects, and more general three-dimensional magnetic actuation mechanisms are not considered.

6.4 Future Work

Several directions for future research emerge from the present work.

A natural extension is the development of a fully three-dimensional formulation for magnetoelastic rods constrained within spatial channels. Such an extension would enable treatment of realistic insertion pathways involving torsion, out-of-plane deformation, and spatially varying curvature.

Incorporation of dynamic effects would allow investigation of transient insertion processes, including inertial response and time-dependent actuation.

Relaxation of the ideal contact assumptions presents another important direction. Incorporating friction and deformable channel walls would provide a more realistic representation of practical confined insertion systems.

Extension to fully three-dimensional magnetoelastic rod models would enable analysis of torsion, spatial magnetic actuation, and navigation through general three-dimensional pathways. The magnetic actuation model may be generalized to include non-uniform magnetic fields, spatially varying field gradients, and more complex magnetization distributions. Such extensions would broaden the range of achievable actuation strategies and improve practical controllability.

Experimental validation remains a critical next step. Physical realization of magnetoelastic rod insertion experiments would enable assessment of the theoretical predictions, particularly the elimination of lateral contact forces and the feasibility of

constant-magnetization insertion strategies in confined geometries.

Finally, the biologically inspired channel geometries considered in this work suggest possible applications in navigation through complex confined pathways, motivating further exploration toward minimally invasive robotic systems, catheter guidance, and magnetically actuated soft robotic devices.

Appendices

Appendix A

Variational Derivation of Equilibrium Equations and Boundary Conditions

Problem Statement: To find the equilibrium configuration of a magnetoelastic rod that is inextensible, unshearable, confined in a given planar curved channel, and acted upon by an external force $\mathbf{F}_{ext}$ at its left end.

$$\min_{\mathbf{r}(s), \theta(s)} \Pi = \int_{s'=0}^L \left\{ \frac{1}{2} EI \theta_s^2(s') - A \mathbf{m}(s') \cdot \mathbf{H}_{ext} \right\} ds' - \mathbf{F}_{ext} \cdot \mathbf{r}(0)$$

Subject to :

$$\nu(s) - 1 = 0$$

$$\eta(s) = 0$$

$$\phi_\Gamma(\mathbf{r}) = 0$$

The augmented Lagrangian is given by

$$\mathcal{L} = \int_0^L \left\{ \underbrace{\frac{1}{2} EI \theta_s^2}_{\text{Bending Energy}} + \underbrace{-A \mathbf{m} \cdot \mathbf{H}_{ext}}_{\text{Zeeman Energy}} + \underbrace{N \{\nu - 1\}}_{\text{Inextensibility}} + \underbrace{H \eta}_{\text{Unshearability}} + \underbrace{f \phi_\Gamma(\mathbf{r})}_{\text{Channel Constraint}} \right\} ds - \underbrace{\mathbf{F}_{ext} \cdot \mathbf{r}(0)}_{\text{Work Done by External Force}}$$

where $N(s)$, $H(s)$, $f(s)$ are the Lagrange parameters that represent the external forces acting on the rod.

A solution $(\mathbf{r}, \theta)$ must satisfy $\delta\mathcal{L} = 0$ for all variations defined on $[0, L]$. Now,

$$\begin{aligned}\delta\mathcal{L} &= \frac{d}{d\epsilon}[\mathcal{L}[\mathbf{r} + \epsilon\delta\mathbf{r}, \theta + \epsilon\delta\theta, N + \epsilon\delta N, H + \epsilon\delta H, f + \epsilon\delta f]] \Big|_{\epsilon=0} = \frac{d}{d\epsilon}\mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] \Big|_{\epsilon=0} \\ &= \frac{d}{d\epsilon}\mathcal{L} \Big|_{\epsilon=0}\end{aligned}$$

Now,

$$\begin{aligned}\mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] &= \int_{s=0}^L \left\{ \frac{1}{2}EI(\theta_s + \epsilon\delta\theta_s)^2 \right. \\ &\quad - A \left[M_s [\cos(\theta + \epsilon\delta\theta + \gamma) \mathbf{e}_1 + \sin(\theta + \epsilon\delta\theta + \gamma) \mathbf{e}_2] \right] \cdot \mathbf{H}_{ext} \\ &\quad + \left[N + \epsilon\delta N \right] \{ (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [\cos(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \sin(\theta + \epsilon\delta\theta) \mathbf{e}_2] - 1 \} \\ &\quad + \left[H + \epsilon\delta H \right] (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [-\sin(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \cos(\theta + \epsilon\delta\theta) \mathbf{e}_2] \\ &\quad \left. + (f + \epsilon\delta f) \phi_\Gamma(\mathbf{r} + \epsilon\delta\mathbf{r}) \right\} ds - \mathbf{F}_{ext} \cdot (\mathbf{r} + \epsilon\delta\mathbf{r})(0).\end{aligned}$$

Differentiating w.r.t. ϵ , we get

$$\begin{aligned}\frac{d}{d\epsilon}\mathcal{L} &= \int_0^L \left\{ EI(\theta_s + \epsilon\delta\theta_s)\delta\theta_s \right. \\ &\quad - A \left[M_s [-\sin(\theta + \epsilon\delta\theta + \gamma) \mathbf{e}_1 + \cos(\theta + \epsilon\delta\theta + \gamma) \mathbf{e}_2] \right] \cdot \mathbf{H}_{ext} \delta\theta \\ &\quad + \delta N \{ (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [\cos(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \sin(\theta + \epsilon\delta\theta) \mathbf{e}_2] - 1 \} \\ &\quad + \left[N + \epsilon\delta N \right] \{ \delta\mathbf{r}_s \cdot [\cos(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \sin(\theta + \epsilon\delta\theta) \mathbf{e}_2] \} \\ &\quad + \left[N + \epsilon\delta N \right] \{ (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [-\sin(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \cos(\theta + \epsilon\delta\theta) \mathbf{e}_2] \delta\theta \} \\ &\quad + \delta H (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [-\sin(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \cos(\theta + \epsilon\delta\theta) \mathbf{e}_2] \\ &\quad + \left[H + \epsilon\delta H \right] \delta\mathbf{r}_s \cdot [-\sin(\theta + \epsilon\delta\theta) \mathbf{e}_1 + \cos(\theta + \epsilon\delta\theta) \mathbf{e}_2] \\ &\quad + \left[H + \epsilon\delta H \right] (\mathbf{r}_s + \epsilon\delta\mathbf{r}_s) \cdot [-\cos(\theta + \epsilon\delta\theta) \mathbf{e}_1 - \sin(\theta + \epsilon\delta\theta) \mathbf{e}_2] \delta\theta \\ &\quad \left. + \delta f \phi_\Gamma(\mathbf{r} + \epsilon\delta\mathbf{r}) + (f + \epsilon\delta f) \nabla \phi_\Gamma(\mathbf{r} + \epsilon\delta\mathbf{r}) \cdot \delta\mathbf{r} \right\} ds - \mathbf{F}_{ext} \cdot \delta\mathbf{r}(0).\end{aligned}$$

Therefore,

$$\begin{aligned} \left. \frac{d}{d\epsilon} \mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] \right|_{\epsilon=0} &= \int_0^L \left\{ EI(\theta_s) \delta\theta_s \right. \\ &\quad - A \left[M_s [-\sin(\theta + \gamma) \mathbf{e}_1 + \cos(\theta + \gamma) \mathbf{e}_2] \right] \cdot \mathbf{H}_{ext} \delta\theta \\ &\quad + \delta N \{ (\mathbf{r}_s) \cdot [\cos(\theta) \mathbf{e}_1 + \sin(\theta) \mathbf{e}_2] - 1 \} \\ &\quad + [N] \{ \delta \mathbf{r}_s \cdot [\cos(\theta) \mathbf{e}_1 + \sin(\theta) \mathbf{e}_2] \} \\ &\quad + [N] \{ (\mathbf{r}_s) \cdot [-\sin(\theta) \mathbf{e}_1 + \cos(\theta) \mathbf{e}_2] \delta\theta \} \\ &\quad + \delta H(\mathbf{r}_s) \cdot [-\sin(\theta) \mathbf{e}_1 + \cos(\theta) \mathbf{e}_2] \\ &\quad + [H] \delta \mathbf{r}_s \cdot [-\sin(\theta) \mathbf{e}_1 + \cos(\theta) \mathbf{e}_2] \\ &\quad + [H] (\mathbf{r}_s) \cdot [-\cos(\theta) \mathbf{e}_1 - \sin(\theta) \mathbf{e}_2] \delta\theta \\ &\quad \left. + \delta f \phi_\Gamma(\mathbf{r}) + (f) \nabla \phi_\Gamma(\mathbf{r}) \cdot \delta \mathbf{r} \right\} ds - \mathbf{F}_{ext} \cdot \delta \mathbf{r}(0). \end{aligned}$$

Simplifying,

$$\begin{aligned} \left. \frac{d}{d\epsilon} \mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] \right|_{\epsilon=0} &= \int_0^L \left\{ EI\theta_s \delta\theta_s - A[\mathbf{m}(s) \times \mathbf{H}_{ext}] \cdot \mathbf{e}_2 \delta\theta \right. \\ &\quad + \delta N (\mathbf{r}_s \cdot \mathbf{d}_3 - 1) + N(\mathbf{d}_3 \cdot \delta \mathbf{r}_s) + N(\mathbf{r}_s \cdot \mathbf{d}_1) \delta\theta \\ &\quad + \delta H \mathbf{r}_s \cdot \mathbf{d}_1 + H \mathbf{d}_1 \cdot \delta \mathbf{r}_s - H (\mathbf{r}_s \cdot \mathbf{d}_3) \delta\theta \\ &\quad \left. + \delta f \phi_\Gamma(\mathbf{r}) + f \nabla \phi_\Gamma(\mathbf{r}) \cdot \delta \mathbf{r} \right\} ds - \mathbf{F}_{ext} \cdot \delta \mathbf{r}(0). \end{aligned}$$

Simplifying further, we get

$$\begin{aligned} \left. \frac{d}{d\epsilon} \mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] \right|_{\epsilon=0} &= \int_0^L \left\{ EI\theta_s \delta\theta_s - A[\mathbf{m}(s) \times \mathbf{H}_{ext}] \cdot \mathbf{e}_3 \delta\theta \right. \\ &\quad + \delta N (\nu - 1) + N \mathbf{d}_3 \cdot \delta \mathbf{r}_s + N \eta \delta\theta \\ &\quad + \delta H \eta + H \mathbf{d}_1 \cdot \delta \mathbf{r}_s - H \nu \delta\theta \\ &\quad \left. + \delta f \phi_\Gamma(\mathbf{r}) + f \nabla \phi_\Gamma(\mathbf{r}) \cdot \delta \mathbf{r} \right\} ds - \mathbf{F}_{ext} \cdot \delta \mathbf{r}(0). \end{aligned}$$

Or,

$$\begin{aligned} \delta\mathcal{L} = \frac{d}{d\epsilon}\mathcal{L}[\mathbf{r}_\epsilon, \theta_\epsilon, N_\epsilon, H_\epsilon, f_\epsilon] \Big|_{\epsilon=0} &= \int_0^L \left\{ EI\theta_s \delta\theta_s - A[\mathbf{m}(s) \times \mathbf{H}_{ext}] \cdot \mathbf{e}_3 \delta\theta \right. \\ &\quad - (H\nu - N\eta) \delta\theta + (N\mathbf{d}_3 + H\mathbf{d}_1) \cdot \delta\mathbf{r}_s + f\nabla\phi_\Gamma(\mathbf{r}) \cdot \delta\mathbf{r} \\ &\quad \left. + \delta N \{\nu - 1\} + \delta H\eta + \delta f\phi_\Gamma(\mathbf{r}) \right\} ds - \mathbf{F}_{ext} \cdot \delta\mathbf{r}(0). \end{aligned}$$

We know that

$$\begin{aligned} \frac{d}{ds} \{(N\mathbf{d}_3 + H\mathbf{d}_1) \cdot \delta\mathbf{r}\} &= \frac{d}{ds}(\mathbf{n} \cdot \delta\mathbf{r}) = \mathbf{n}_s \cdot \delta\mathbf{r} + \mathbf{n} \cdot \delta\mathbf{r}_s, \\ \frac{d}{ds}(\theta_s \delta\theta) &= \theta_{ss} \delta\theta + \theta_s \delta\theta_s \text{ and} \\ H\nu - N\eta &= (\mathbf{r}_s \times \mathbf{n}) \cdot \mathbf{e}_3 \end{aligned}$$

Replacing the expression for $\mathbf{n} \cdot \delta\mathbf{r}_s$, $B\theta_s \delta\theta_s$ and $N\eta - H\nu$ into the above expression, we get

$$\begin{aligned} \delta\mathcal{L} &= \int_0^L \left\{ -\mathbf{n}_s + f\nabla\phi_\Gamma(\mathbf{r}) \right\} \cdot \delta\mathbf{r} ds + \int_0^L \left\{ -EI\theta_{ss}\mathbf{e}_3 - A\mathbf{m}(s) \times \mathbf{H}_{ext} - \mathbf{r}_s \times \mathbf{n} \right\} \delta\theta ds \\ &\quad + \int_0^L (\nu - 1)\delta N ds + \int_0^L \eta \delta H ds + \int_0^L \phi_\Gamma(\mathbf{r}) \delta f ds - \mathbf{F}_{ext} \cdot \delta\mathbf{r}(0) + \mathbf{n} \cdot \delta\mathbf{r} \Big|_0^L + B\theta_s \delta\theta \Big|_0^L = 0. \end{aligned}$$

From the fundamental lemma of the calculus of variations for $s \in (0, L)$, we get

$$\begin{aligned} &\left. \begin{aligned} \mathbf{n}_s - f\nabla\phi(\mathbf{r}) &= \mathbf{0}, \\ EI\theta_{ss}\mathbf{e}_3 + A\mathbf{m}(s) \times \mathbf{H}_{ext} + \mathbf{r}_s \times \mathbf{n} &= \mathbf{0}, \end{aligned} \right\} \text{equilibrium equations} \\ &\nu = 1, \\ &\eta = 0, \text{ and} \\ &\phi_\Gamma(\mathbf{r}) = 0. \end{aligned}$$

$s \in (0, L)$

From $\nu(s) = 1$ and $\eta(s) = 0$, we get

$$\mathbf{r}_s(s) = \mathbf{d}_3(s) \quad \text{and} \quad |\mathbf{r}(s) \times \mathbf{n}(s)| = H(s).$$

From $\phi_\Gamma(\mathbf{r}) = 0$, we get $\mathbf{r}(s) = \mathbf{\Gamma}(\sigma(s))$. Since $\mathbf{\Gamma}(s)$ is arc length parameterized, $|\mathbf{\Gamma}_s(s)| = 1$. This implies $|\sigma_s(s)| = 1$. since the rod is moving inside the channel, $\sigma_s(s) = 1$. Thus, $\sigma(s) = s + s_0$. Therefore,

$$\mathbf{r}(s) = \mathbf{\Gamma}(s + s_0) \quad \text{and} \quad \theta(s) = \Theta(s_0 + s).$$

Let $k_c(s_0 + s) = \Theta_s(s_0 + s)$ and $\kappa(s) = \theta_s(s)$. Then

$$\kappa(s) = k_c(s_0 + s).$$

$$s \in \{0, L\}$$

Enforcing the constraint $\phi_\Gamma(\mathbf{r}) = 0$ at $s = 0$ and $s = L$, we get $\mathbf{r}(0) = \mathbf{\Gamma}(s_0)$, $\mathbf{r}(L) = \mathbf{\Gamma}(s_L)$, $\theta(0) = \Theta(s_0)$ and $\theta(L) = \Theta(s_L)$. Now,

$$\begin{aligned} \delta \mathbf{r}(0) &= \frac{d\mathbf{\Gamma}}{ds_0} \delta s_0 = \mathbf{\Gamma}_s(s_0) \delta s_0 = \mathbf{r}_s(0) \delta s_0 = \mathbf{d}_3(0) \delta s_0, \\ \delta \theta(0) &= \frac{d\Theta}{ds_0} \delta s_0 = \Theta_s(s_0) \delta s_0 = \theta_s(s_0) \delta s_0 = \kappa(s_0) \delta s_0, \\ \delta \mathbf{r}(L) &= \frac{d\mathbf{\Gamma}}{ds_L} \delta s_L = \mathbf{\Gamma}_s(s_L) \delta s_L = \mathbf{r}_s(L) \delta s_L = \mathbf{d}_3(L) \delta s_L, \\ \delta \theta(L) &= \frac{d\Theta}{ds_L} \delta s_L = \Theta_s(s_L) \delta s_L = \theta_s(s_L) \delta s_L = \kappa(s_L) \delta s_L. \end{aligned}$$

Now, we have

$$-\mathbf{F}_{ext} \cdot \delta \mathbf{r}(0) + \mathbf{n} \cdot \delta \mathbf{r} \Big|_0^L + EI \theta_s \delta \theta \Big|_0^L = 0.$$

Putting the above-derived expressions together, we get

$$-\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) \delta s_0 + \mathbf{n}(L) \cdot \mathbf{\Gamma}_s(s_L) \delta s_L - \mathbf{n}(0) \cdot \mathbf{\Gamma}_s(s_0) \delta s_0 + EI \kappa^2(L) \delta s_L - EI \kappa^2(0) \delta s_0 = 0.$$

Thus, we have

$$\delta s_0 \left[-\mathbf{n}(0) \cdot \mathbf{\Gamma}_s(s_0) - \mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI \kappa^2(0) \right] - \delta s_L \left[-\mathbf{n}(L) \cdot \mathbf{\Gamma}_s(s_L) - EI \kappa^2(L) \right] = 0.$$

Since this is a force-controlled problem i.e. $\mathbf{F}_{ext}$ is specified, $\delta s_0 \neq 0$ and $\delta s_L \neq 0$. Therefore, we have

$$N(0) + \mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) + EI \kappa^2(0) = 0$$

$$N(L) + EI\kappa^2(L) = 0.$$

Appendix B

Expression for Internal Forces and Lateral Force in a Continuous Curvature Channel

Taking the components of the equilibrium equations in the director frame, we get

$$\begin{aligned}N_s - \kappa H &= 0 \\ \kappa N + H_s - f &= 0 \\ EI\kappa_s + H + C\tau &= 0\end{aligned}$$

Thus, we have

$$f = \kappa N + H_s$$

where

$$\begin{aligned}N &= N(0) + \int_0^s \kappa(s')H(s') \, ds' \\ &= -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI\kappa^2(0) + \int_0^s \kappa(s')H(s') \, ds' \\ H_s &= -EI\kappa_{ss} - C\tau_s\end{aligned}$$

Putting the above expressions into f , we get

$$f(s) = -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0)\kappa - EI\kappa_{ss} - EI\kappa\kappa^2(0) - EI\kappa \int_0^s \kappa\kappa_s \, ds' - C\kappa \int_0^s \kappa(s')\tau(s') \, ds' - C\tau_s$$

In the above equations, $\kappa = \kappa(s)$. We have

$$N(s) = N(0) + \int_0^s \kappa(s') H(s') ds'.$$

Substituting $s = L$, we get

$$\begin{aligned} N(L) - N(0) &= \int_0^L \kappa(s') H(s') ds' \\ &= \int_0^L \kappa \left\{ -EI\kappa_s - C\tau \right\} ds' \\ &= -\frac{EI}{2} \int_0^L 2\kappa\kappa_s ds' - C \int_0^L \kappa(s')\tau(s') ds' \\ &= -\frac{EI}{2} \kappa^2(s) \Big|_0^L - C \int_0^L \kappa(s')\tau(s') ds' \end{aligned}$$

From the boundary conditions, we have

$$N(L) - N(0) = -EI\kappa^2(L) + EI\kappa^2(0) + \mathbf{F}_{ext} \cdot \Gamma_s(s_0) = -EI\kappa^2(s) \Big|_0^L + \mathbf{F}_{ext} \cdot \Gamma_s(s_0).$$

Equating the two expressions, we get

$$\mathbf{F}_{ext} \cdot \Gamma_s(s_0) = \frac{EI}{2} \kappa^2(L) - \frac{EI}{2} \kappa^2(0) - C \int_0^L \kappa(s')\tau(s') ds'$$

Thus,

$$f(s) = -\frac{EI}{2} \kappa(s)\kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI\kappa_{ss}(s) + C\kappa(s) \int_s^L \kappa(s')\tau(s') ds' - C\tau_s(s).$$

Putting $\tau(s) = 0$ yields the expressions for the case of an elastic rod.

Appendix C

General Solution for Differential Equation

We have

$$\kappa(s)\tau_{ss} - \kappa_s(s)\tau_s(s) + \kappa^3(s)\tau(s) - \psi_\Gamma(s) = 0$$

where

$$\psi_\Gamma(s) = \frac{1}{C} \left(\Lambda_{\Gamma,s}(s)\kappa(s) - \kappa_s(s)\Lambda_\Gamma(s) \right).$$

Simplifying the expression for $\psi_\Gamma(s)$, we get

$$\psi_\Gamma(s) = -\frac{EI}{C} \left(\kappa\kappa_{sss} - \kappa_s\kappa_{ss} + \kappa^3\kappa_s \right).$$

Thus, the differential equation simplifies to

$$\kappa\tau_{ss} - \kappa_s\tau_s + \kappa^3\tau = -\frac{EI}{C} \left(\kappa\kappa_{sss} - \kappa_s\kappa_{ss} + \kappa^3\kappa_s \right) = h(s).$$

Let

$$t(s) = \theta(s) - \theta(0).$$

Then

$$\begin{aligned} \tau_s &= \frac{d\tau}{ds} = \frac{d\tau}{dt} \frac{dt}{ds} = \kappa\tau_t, \quad \text{and} \\ \tau_{ss} &= \kappa_s\tau_t + \kappa \frac{d\tau_t}{ds} = \kappa_s\tau_t + \kappa^2\tau_{tt}. \end{aligned}$$

Substituting the above expression into the differential equation, we get

$$\begin{aligned}\tau_{tt} + \tau &= -\frac{EI}{C} \left(t_{ss} + \frac{t_s t_{ssss} - t_{ss} t_{sss}}{t_s^3} \right). \\ \implies \tau_{tt} + \tau &= h(t).\end{aligned}$$

This gives the general solution as

$$\tau(t) = \tau_h(t) + \tau_p(t),$$

where

$$\tau_h(t) = A_1 \sin t + B_1 \cos t.$$

Let $\tau_p(t) = u_1(t) \sin t + u_2(t) \cos t$. Then, imposing the constraints gives the following equations

$$\begin{bmatrix} \sin t & \cos t \\ \cos t & -\sin t \end{bmatrix} \begin{bmatrix} u_{1,t} \\ u_{2,t} \end{bmatrix} = \begin{bmatrix} 0 \\ h(t) \end{bmatrix}$$

Thus, we have

$$\begin{aligned}u_1(t) &= \int_0^t h(t') \cos t' dt' \\ u_2(t) &= -\int_0^t h(t') \sin t' dt'\end{aligned}$$

Therefore,

$$\tau_p(t) = \int_0^t h(t') \sin(t - t') dt'$$

Let $t' = t(s')$. Then $dt' = k(s') ds'$. Then,

$$\tau_p(s) = \int_0^s h(s') k(s') \sin [\theta(s) - \theta(s')] ds'.$$

Thus,

$$\tau(s) = A_1 \sin [\theta(s) - \theta(0)] + B_1 \cos [\theta(s) - \theta(0)] - \frac{EI}{C} \int_0^s g(s') \sin [\theta(s) - \theta(s')] ds',$$

where

$$g(s) = \frac{d}{ds} \left(\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} \right).$$

Now,

$$\tau_{p,s}(s) = \frac{EI}{C} k(s) \int_0^s g(s') \cos [\theta(s) - \theta(s')] ds'.$$

Using the boundary conditions, we get

$$A_1 = \tau_0 \tan [\theta(L) - \theta(0)] + \frac{\tau_s(L) - \tau_{p,s}(L)}{\kappa(L) \cos [\theta(L) - \theta(0)]}$$

$$B_1 = \tau_0$$

Spiral

For a spiral channel, $\kappa(s) = Ks$ and $\Lambda_\Gamma(s) = -\frac{BK^3s(L^2+s^2)}{2}$. The general solution is

$$\tau(s) = A_1 \sin \frac{Ks^2}{2} + B_1 \cos \frac{Ks^2}{2} + \tau_p(s)$$

Now,

$$\begin{aligned} \tau_p(t) &= \int_0^t h(t') \sin(t - t') dt' = \int_0^t -\frac{EIK}{C} \sin(t - t') dt' \\ &= -\frac{EIK}{C} (1 - \cos t) \end{aligned}$$

Thus,

$$\tau_p(s) = -\frac{EIK}{C} \left(1 - \cos \frac{Ks^2}{2} \right)$$

Using the boundary conditions, we get

$$A_1 = \tau_0 \tan \frac{KL^2}{2} - \frac{EIK^2L^2}{C \cos \frac{KL^2}{2}} - \frac{EIK}{C} \tan \frac{KL^2}{2}$$

$$B_1 = \tau_0$$

Sinusoidal

For a sinusoidal channel, $\kappa(s) = X \sin(\frac{s}{Y})$ and

$$\Lambda_{\Gamma}(s) = -\frac{EIX^3}{2} \sin^2\left(\frac{L}{Y}\right) \sin\left(\frac{s}{Y}\right) - \frac{EIX^3}{2} \sin^3\left(\frac{s}{Y}\right) + \frac{EIX}{Y^2} \sin\left(\frac{s}{Y}\right).$$

Thus, $\theta(s) = -XY \cos(\frac{s}{Y}) + XY + \theta(0)$. The general solution is

$$\tau(s) = A_1 \sin \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] + B_1 \cos \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] + \tau_p(s)$$

Now,

$$\begin{aligned} \tau_p(t) &= \int_0^t h(t') \sin(t - t') dt' \\ &= \int_0^t \left[\frac{EI}{CY^2} t' + \frac{B\theta(0)}{CY^2} \right] \sin(t - t') dt' \\ &= -\frac{EI}{CY^2} (t + \sin t) - \frac{EI\theta(0)}{CY^2} (1 - \cos t) \end{aligned}$$

Thus,

$$\begin{aligned} \tau_p(s) &= -\frac{EI}{CY^2} \left(-XY \cos\left(\frac{s}{Y}\right) + XY + \sin \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] \right) \\ &\quad - \frac{EI\theta(0)}{CY^2} \left(1 - \cos \left[-XY \cos\left(\frac{s}{Y}\right) + XY \right] \right) \end{aligned}$$

We can obtain the expressions for A_1 and B_1 [3.22](#) and [3.23](#).

Appendix D

Jump Conditions at Discontinuity

Elastic Rod

Jump in $N(s)$ at $s = l$

$$\begin{aligned}\lim_{h \rightarrow 0} \int_{l-h}^{l+h} \{N_s - \kappa H\} ds &= 0 \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} \{N_s - \kappa[-EI\kappa_s]\} ds &= 0 \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} \left\{N_s + \frac{EI}{2} \frac{d(\kappa^2)}{ds}\right\} ds &= 0\end{aligned}$$

$$\text{Thus, } \lim_{h \rightarrow 0} N(l+h) - N(l-h) = \llbracket N(l) \rrbracket = -\frac{EI}{2} \llbracket \kappa^2(l) \rrbracket \rightarrow \text{point load along } \mathbf{d}_3$$

Jump in $H(s)$ at $s = l$

$$\begin{aligned}H(s) &= -B\kappa_s(s) = 0 \\ \lim_{h \rightarrow 0} H(l+h) - H(l-h) &= \llbracket H(l) \rrbracket = -EI \llbracket \kappa_s(l) \rrbracket \rightarrow \text{point load along } \mathbf{d}_1\end{aligned}$$

Jump in $M(s)$ at $s = l$

From geometry, we know $M(s) = EI\kappa(s)$. Thus,

$$\lim_{h \rightarrow 0} M(l+h) - M(l-h) = \llbracket M(l) \rrbracket = EI \llbracket \kappa(l) \rrbracket \rightarrow \text{concentrated moment along } \mathbf{d}_2$$

Magnetoelastic Rod

Jump in $N(s)$ at $s = l$

$$\begin{aligned}\lim_{h \rightarrow 0} \int_{l-h}^{l+h} \left\{ N_s(s') - \kappa(s') H(s') \right\} ds' &= 0 \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} \left\{ N_s(s') - \kappa(s') \left[-EI\kappa_s(s') - C\tau(s') \right] \right\} ds' &= 0 \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} \left\{ N_s(s') + \frac{EI}{2} \frac{d(\kappa^2)}{ds'} + C\kappa(s')\tau(s') \right\} ds' &= 0\end{aligned}$$

Assuming $\tau(s)$ to atmost have a finite jump discontinuity at $s = l$, we get

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} C\kappa(s')\tau(s') ds' = 0.$$

Thus,

$$\lim_{h \rightarrow 0} N(l+h) - N(l-h) = \llbracket N(l) \rrbracket = -\frac{EI}{2} \llbracket \kappa^2(l) \rrbracket \rightarrow \text{point load along } \mathbf{e}_3$$

Jump in $H(s)$ at $s = l$

$$\begin{aligned}H(s) &= -EI\kappa_s(s) - C\tau(s) \\ \lim_{h \rightarrow 0} H(l+h) - H(l-h) &= \llbracket H(l) \rrbracket = -EI\llbracket \kappa_s(l) \rrbracket - C\llbracket \tau(l) \rrbracket \rightarrow \text{point load along } \mathbf{e}_1\end{aligned}$$

Jump in $M(s)$ at $s = l$

From geometry, we know $M(s) = EI\kappa(s)$. Thus,

$$\begin{aligned}M_s + H + C\tau &= 0 \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} \{M_s(s') + H(s') + C\tau(s')\} ds' &= 0\end{aligned}$$

Since there is only a finite jump in $H(s)$ and $\tau(s)$,

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} H(s') \, ds' = 0$$

$$\lim_{h \rightarrow 0} \int_{l-h}^{l+h} \tau(s') \, ds' = 0$$

Therefore,

$$\lim_{h \rightarrow 0} M(l+h) - M(l-h) = \llbracket M(l) \rrbracket = EI \llbracket \kappa(l) \rrbracket \quad \rightarrow \text{concentrated moment along } \mathbf{e}_2$$

Appendix E

Expression for Lateral Force in a Discontinuous Curvature Channel

Elastic Rod

From the boundary conditions, we have

$$N(L) - N(0) = -EI\kappa^2(L) + EI\kappa^2(0) + \mathbf{F}_{ext} \cdot \Gamma_s(s_0) = -EI\kappa^2(s) \Big|_0^L + \mathbf{F}_{ext} \cdot \Gamma_s(s_0).$$

Now,

$$N(s) = N(0) + \int_0^s \kappa(s')H(s') \, ds'.$$

Putting $s = L$, we get

$$\begin{aligned} N(L) - N(0) &= \int_0^L \kappa(s')H(s') \, ds' \\ &= \int_0^L \kappa(s') \{ -EI\kappa_s \} \, ds' \\ &= -\frac{EI}{2} \int_0^L 2\kappa(s')\kappa_s(s') \, ds' \\ &= -\frac{EI}{2} \left[\int_0^{l-h} \frac{d(\kappa^2)}{ds'} \, ds' + \int_{l-h}^{l+h} \frac{d(\kappa^2)}{ds'} \, ds' + \int_{l+h}^L \frac{d(\kappa^2)}{ds'} \, ds' \right] \\ &= \frac{EI}{2} \kappa^2(0) - \frac{EI}{2} \kappa^2(L). \end{aligned}$$

Thus,

$$\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) = \frac{EI}{2} \kappa^2(L) - \frac{EI}{2} \kappa^2(0).$$

Using equilibrium equations, we have

$$f = \kappa N + H_s,$$

where

$$\begin{aligned} N &= N(0) + \int_0^s \kappa(s') H(s') \, ds' \\ &= -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI \kappa^2(0) + \int_0^s \kappa(s') H(s') \, ds', \\ H_s &= -EI \kappa_{ss}. \end{aligned}$$

Putting the above expressions into f , we get

$$\begin{aligned} f(s) &= \kappa(s) \left\{ -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI \kappa^2(0) \right\} + \kappa \int_0^s \kappa(s') \{ -EI \kappa_s(s') \} \, ds' - EI \kappa_{ss} \\ &= -\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) \kappa(s) - EI \kappa(s) \kappa^2(0) - EI \kappa(s) \int_0^s \kappa(s') \kappa_s(s') \, ds' - EI \kappa_{ss}(s) \\ &= -\frac{EI}{2} \kappa(s) \kappa^2(L) - \frac{EI}{2} \kappa^3(s) - EI \kappa_{ss}(s). \end{aligned}$$

Note that

$$\begin{aligned} \kappa(s) &= \kappa_1(s) + \left(\kappa_2(s) - \kappa_1(s) \right) H(s-l), \\ \kappa_{ss}(s) &= \kappa_{1,ss}(s) + \left(\kappa_{2,ss}(s) - \kappa_{1,ss}(s) \right) H(s-l) + \llbracket \kappa_s(l) \rrbracket \delta(s-l) + \llbracket \kappa(l) \rrbracket \delta'(s-l). \end{aligned}$$

Also,

$$\begin{aligned} \lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s') \, ds' &= -EI \llbracket \kappa_s(l) \rrbracket \quad \rightarrow \text{point load along } \mathbf{d}_1, \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s') (s' - l) \, ds' &= EI \llbracket \kappa(l) \rrbracket \quad \rightarrow \text{concentrated moment along } \mathbf{d}_2. \end{aligned}$$

Magnetoelastic Rod

From the boundary conditions, we have

$$N(L) - N(0) = -EI\kappa^2(L) + EI\kappa^2(0) + \mathbf{F}_{ext} \cdot \Gamma_s(s_0) = -EI\kappa^2(s) \Big|_0^L + \mathbf{F}_{ext} \cdot \Gamma_s(s_0).$$

Now,

$$N(s) = N(0) + \int_0^s \kappa(s')H(s') \, ds'.$$

Putting $s = L$, we get

$$\begin{aligned} N(L) - N(0) &= \int_0^L \kappa(s')H(s') \, ds' \\ &= \int_0^L \kappa(s') \{ -EI\kappa_s(s') - C\tau(s') \} \, ds' \\ &= -\frac{EI}{2} \int_0^L 2\kappa(s')\kappa_s(s') \, ds' - C \int_0^L \kappa(s')\tau(s') \, ds' \\ &= -\frac{EI}{2} \left[\int_0^{l-h} \frac{d(\kappa^2)}{ds'} \, ds' + \int_{l-h}^{l+h} \frac{d(\kappa^2)}{ds'} \, ds' + \int_{l+h}^L \frac{d(\kappa^2)'}{ds} \, ds' \right] \\ &\quad - C \int_0^L \kappa(s')\tau(s') \, ds' \\ &= -\frac{EI}{2} \kappa^2(s) \Big|_0^L - C \int_0^L \kappa(s')\tau(s') \, ds'. \end{aligned}$$

Thus,

$$\mathbf{F}_{ext} \cdot \Gamma_s(s_0) = \frac{EI}{2} \kappa^2(s) \Big|_0^L - C \int_0^L \kappa(s')\tau(s') \, ds' \quad \rightarrow \text{same as continuous curvature}$$

We have

$$f = \kappa N + H_s$$

where

$$\begin{aligned} N &= N(0) + \int_0^s \kappa(s')H(s') \, ds' \\ &= -\mathbf{F}_{ext} \cdot \Gamma_s(s_0) - EI\kappa^2(0) + \int_0^s \kappa(s')H(s') \, ds', \end{aligned}$$

$$H_s = -EI\kappa_{ss} - C\tau.$$

Putting the above expressions into f , we get

$$f(s) = \kappa[-\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0) - EI\kappa^2(0)] + \kappa \int_0^s \kappa(s')[-EI\kappa_s(s') - C\tau(s')] ds' - Bk_{ss} - C\tau_s.$$

Substituting the expression for $\mathbf{F}_{ext} \cdot \mathbf{\Gamma}_s(s_0)$ and simplifying, we get

$$f(s) = -\frac{EI}{2}\kappa(s)\kappa^2(L) - \frac{EI}{2}\kappa^3(s) - EI\kappa_{ss}(s) + C\kappa(s) \int_s^L \kappa(s')\tau(s') ds' - C\tau_s(s).$$

Note that

$$\begin{aligned} \kappa(s) &= \kappa_1(s) + \left(\kappa_2(s) - \kappa_1(s)\right) H(s-l), \\ \kappa_{ss}(s) &= \kappa_{1,ss}(s) + \left(\kappa_{2,ss}(s) - \kappa_{1,ss}(s)\right) H(s-l) + \llbracket \kappa_s(l) \rrbracket \delta(s-l) + \llbracket \kappa(l) \rrbracket \delta'(s-l). \end{aligned}$$

Also,

$$\begin{aligned} \lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s') ds' &= -EI \llbracket \kappa_s(l) \rrbracket - C \llbracket \tau(l) \rrbracket \quad \rightarrow \text{point load along } \mathbf{d}_1, \\ \lim_{h \rightarrow 0} \int_{l-h}^{l+h} f(s')(s' - l) ds' &= EI \llbracket \kappa(l) \rrbracket \quad \rightarrow \text{concentrated moment along } \mathbf{d}_2. \end{aligned}$$

Appendix F

Expression of Channel Curvature for Rod Insertion

Assuming $\kappa(s) \neq 0$, dividing eq. (3.27) by κ^2 we get

$$\begin{aligned} & \frac{d}{ds} \left(\frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} \right) = 0 \\ \implies & \frac{\kappa^2}{2} + \frac{\kappa_{ss}}{\kappa} = C_1 \\ \implies & \frac{\kappa^3 \kappa_s}{2} + \kappa_s \kappa_{ss} = C_1 \kappa \kappa_s \\ \implies & \frac{d}{ds} \left(\frac{\kappa^4}{4} + \kappa_s^2 - C_1 \kappa^2 \right) = 0 \\ \implies & \kappa_s^2 = -\frac{\kappa^4}{4} + C_1 \kappa^2 + C_2 \end{aligned}$$

Let $z = \kappa^2$. Then

$$\kappa_s^2 = P(z), \quad P(z) = -\frac{z^2}{4} + C_1 z + C_2.$$

The roots of $P(z)$ are

$$z_{1,2} = 2C_1 \pm 2\sqrt{C_1^2 + C_2}$$

Define

$$a^2 = 2C_1 + 2\sqrt{C_1^2 + C_2}, \quad b^2 = 2C_1 - 2\sqrt{C_1^2 + C_2},$$

with $a > b \geq 0$. The first integral becomes

$$\kappa_s^2 = \frac{1}{4}(a^2 - \kappa^2)(\kappa^2 - b^2).$$

$$\frac{d\kappa}{\sqrt{(a^2 - \kappa^2)(\kappa^2 - b^2)}} = \pm \frac{1}{2} ds.$$

This integral is of elliptic type. We introduce the substitution

$$\kappa(s) = a \operatorname{cn}(u(s), m),$$

where $\operatorname{cn}(u, m)$ is the *Jacobi elliptic cosine*, defined via

$$\operatorname{cn}(u, m) = \cos(\phi), \quad u = \int_0^\phi \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}}.$$

Thus, $\operatorname{cn}(u, m)$ generalizes the cosine function:

$$\operatorname{cn}(u, 0) = \cos u, \quad \operatorname{cn}(u, 1) = \operatorname{sech} u.$$

Using

$$\frac{d}{du} \operatorname{cn}(u, m) = -\operatorname{sn}(u, m) \operatorname{dn}(u, m),$$

we obtain

$$\kappa_s = -a \operatorname{sn}(u, m) \operatorname{dn}(u, m) u',$$

and hence

$$\kappa_s^2 = a^2 u'^2 \operatorname{sn}^2(u, m) \operatorname{dn}^2(u, m).$$

Using identities:

$$\begin{aligned} \operatorname{sn}^2(u, m) &= 1 - \operatorname{cn}^2(u, m) = \frac{a^2 - \kappa^2}{a^2}, \\ \operatorname{dn}^2(u, m) &= 1 - m \operatorname{sn}^2(u, m) = \frac{(1 - m)a^2 + m\kappa^2}{a^2}, \end{aligned}$$

we obtain

$$\kappa_s^2 = u'^2 \frac{(a^2 - \kappa^2)((1 - m)a^2 + m\kappa^2)}{a^2}.$$

Comparing with

$$\kappa_s^2 = \frac{1}{4}(a^2 - \kappa^2)(\kappa^2 - b^2),$$

and cancelling $(a^2 - k^2)$, we obtain

$$u'^2 \frac{(1-m)a^2 + m\kappa^2}{a^2} = \frac{1}{4}(\kappa^2 - b^2).$$

Equating coefficients of k^2 and constants gives:

$$\begin{aligned} u'^2 \frac{m}{a^2} &= \frac{1}{4}, \\ u'^2(1-m) &= -\frac{1}{4}b^2. \end{aligned}$$

Solving,

$$m = \frac{a^2 - b^2}{a^2}, \quad u' = \frac{a}{2\sqrt{m}}.$$

Thus,

$$\begin{aligned} u(s) &= \frac{a}{2\sqrt{m}}(s - s_c). \\ \kappa(s) &= a \operatorname{cn} \left(\frac{a}{2\sqrt{m}}(s - s_c), m \right) \end{aligned} \tag{F.1}$$

with

$$\begin{aligned} a^2 &= 2C_1 + 2\sqrt{C_1^2 + C_2}, \\ b^2 &= 2C_1 - 2\sqrt{C_1^2 + C_2}, \\ m &= \frac{a^2 - b^2}{a^2}. \end{aligned}$$

Note:

- $C_1 = \frac{a^2 + b^2}{4} \geq 0$.
- $C_2 = -\frac{a^2 b^2}{4} \leq 0$.
- $m = 1$: $\operatorname{cn}(u, 1) = \operatorname{sech}(u)$. Let $\lambda = \sqrt{C_1}$. Thus,

$$\kappa(s) = 2\lambda \operatorname{sech} \left[\lambda(s - s_c) \right].$$

Bibliography

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